

Exercises NMSM 2024-2025

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June 2025

Introduction

This document contains the solution for the exercises of the course Numerical Methods for Soft Matter (year 2024 - 2025) at University of Padova. All the code can be found on GitHub [here](#) together with the images, this document, the source of this document and some animations. Most of the images are vector images, but in some cases (because of million of points) a raster image was put instead.

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Exercises Lecture 1

1.1 Sampling random points within D-dimensional domains by hit and miss

Use the hit-and-miss method to estimate numerically the volume of an ellipsoid of parameters $a = 3$, $b = 2$, $c = 2$ and plot the deviation of the estimate from the analytic value as a function of the number of throws used. Repeat the above procedure to estimate the volume of an ellipsoid of parameters $a = 3$, $b = 1$, $c = 1$. Compare the two results and try to improve the second case if needed.

1.1.1 Solution

By randomly sampling points in the volume $[-a, a] \times [-b, b] \times [-c, c]$ we get the figure 1.1. Each point in the figure is calculated as $|V_{exact} - V_{estimated}|$ where $V_{estimated}$ is the average of 25 runs of hit and miss. At 10^7 points we get:

Exact volume for ellipsoid (3,2,2): 50.2655, best estimate: 50.26699

Exact volume for ellipsoid (3,1,1): 12.5664, best estimate: 12.56602

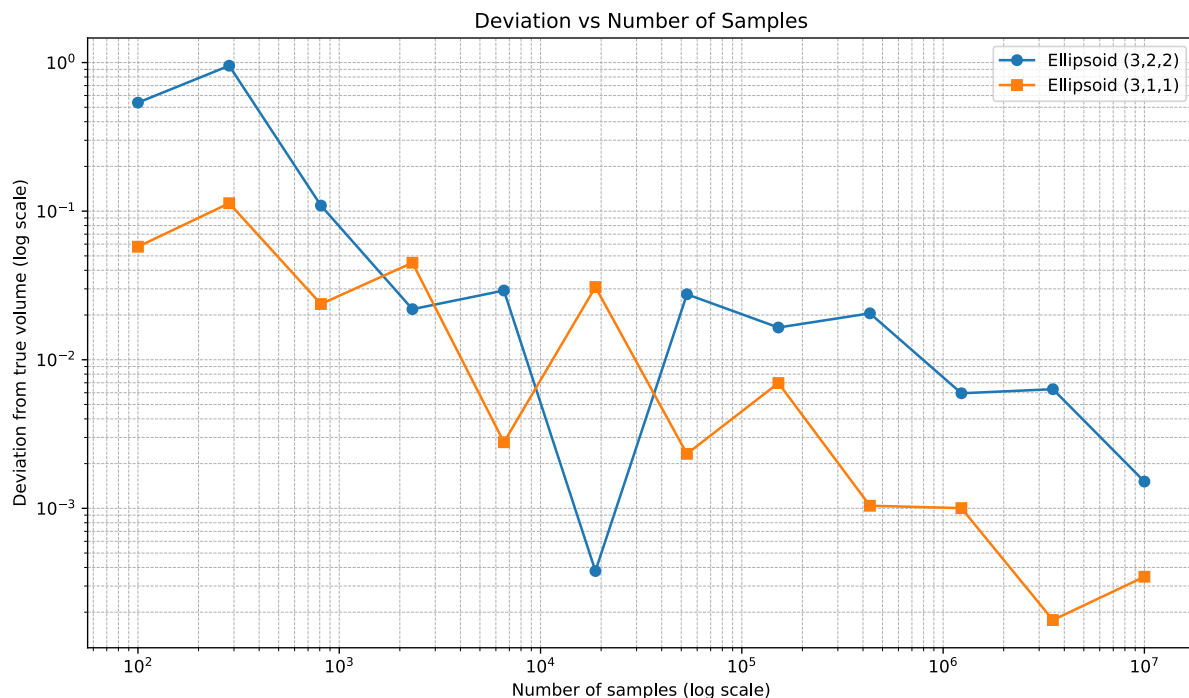


Figure 1.1: Sampling results for the estimate of the volume of an ellipsoid.

1.2 Sampling random numbers from a given distribution: inversion method

Use the inversion method to generate random numbers with the following PDF:

1. $\rho(x) = cxe^{-x^2}$, for $x \in \mathbb{R}^+$.
2. $\rho(x) = bx^4$, for $x \in [0, 3]$.

1.2.1 Solution

First we find the normalization constant by imposing $\int \rho(x)dx = 1$, so we obtain

$$\rho_1(x) = 2xe^{-x^2} \qquad \rho_2(x) = \frac{5}{243}x^4 \qquad (1.1)$$

then we calculate the cumulative $F(x)$:

$$F_1(x) = \int_0^x 2te^{-t^2} dt = 1 - e^{-x^2} \qquad F_2(x) = \int_0^x \frac{5}{243}t^4 dt = \frac{x^5}{243} \qquad (1.2)$$

and finally we obtain $F^{-1}(y)$ by inverting $F(x)$:

$$y = 1 - e^{-x^2} \implies x = F_1^{-1}(y) = \sqrt{\log \frac{1}{1-y}}, \text{ for } y \in [0, 1] \qquad (1.3)$$

$$y = \frac{x^5}{243} \implies x = F_2^{-1}(y) = \sqrt[5]{243y}, \text{ for } y \in [0, 1] \qquad (1.4)$$

By sampling uniformly y and then calculating $F^{-1}(y)$ it's like sampling $\rho(x)$:

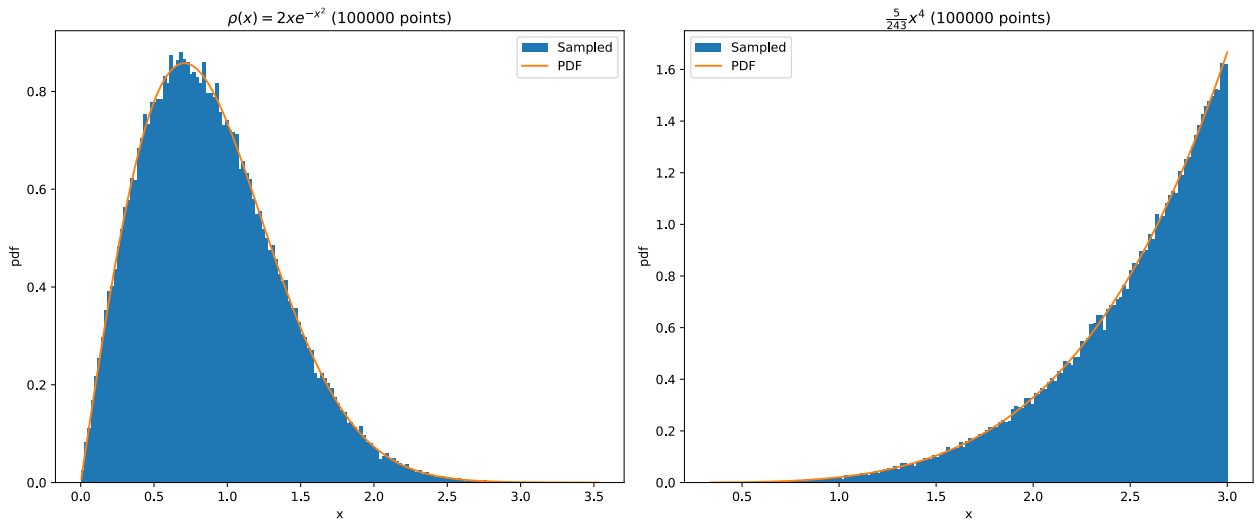


Figure 1.2: Comparison of the sampling with the real pdfs. On the left ρ_1 , on the right ρ_2

Exercises Lecture 2

2.1 Rejection method

Use the rejection method to generate random numbers that are distributed according to the pdf

$$f(x) = A_f e^{-8\left(\frac{x^2}{2} + \frac{x^4}{4}\right)} \quad (2.1)$$

where

$$\int_{-\infty}^{+\infty} f(x) dx = \frac{e}{\sqrt{2}} K_{1/4}(1) \quad (2.2)$$

and $K_{1/4}$ denotes the modified Bessel function of the second kind.

2.1.1 Solution

We assume the A_f is the normalization constant, so $A_f = \frac{\sqrt{2}}{e K_{1/4}(1)}$. Now we need to find an easier PDF $g(x)$ such that $c \cdot g(x) > f(x)$. If we choose $g(x)$ as $\mathcal{N}(0, 0.35)$ we have that:

$$g(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} = \frac{1}{\sqrt{0.245\pi}} e^{-\frac{x^2}{0.245}} \quad (2.3)$$

Now we need to find c such that $c \cdot g(x) > f(x)$ (with c not too far from 1):

$$c \cdot g(x) > f(x) \implies \log \frac{c}{\sqrt{0.245\pi}} - \frac{x^2}{0.245} > \log \frac{\sqrt{2}}{e K_{1/4}(1)} - 8 \left(\frac{x^2}{2} + \frac{x^4}{4} \right) \quad (2.4)$$

by setting $C = \frac{c}{\sqrt{0.245\pi}}$ and $B = \frac{\sqrt{2}}{e K_{1/4}(1)}$ and rearranging the terms:

$$2x^4 + x^2 \left(4 - \frac{1}{0.245} \right) > \log \frac{B}{C} \quad (2.5)$$

since the left term is always positive or zero, then

$$\log \frac{B}{C} < 0 \implies B < C \implies \frac{\sqrt{2}}{e K_{1/4}(1)} < \frac{c}{\sqrt{0.245\pi}} \implies c \gtrsim 1.06 \quad (2.6)$$

so we choose $c = 1.1$. In figure 2.1 we show the difference between the two PDFs.

Now we can apply the Rejection sampling. The results are shown in figure 2.2. The efficiency of the method ($N_{sampled}/N$) is around 91%.

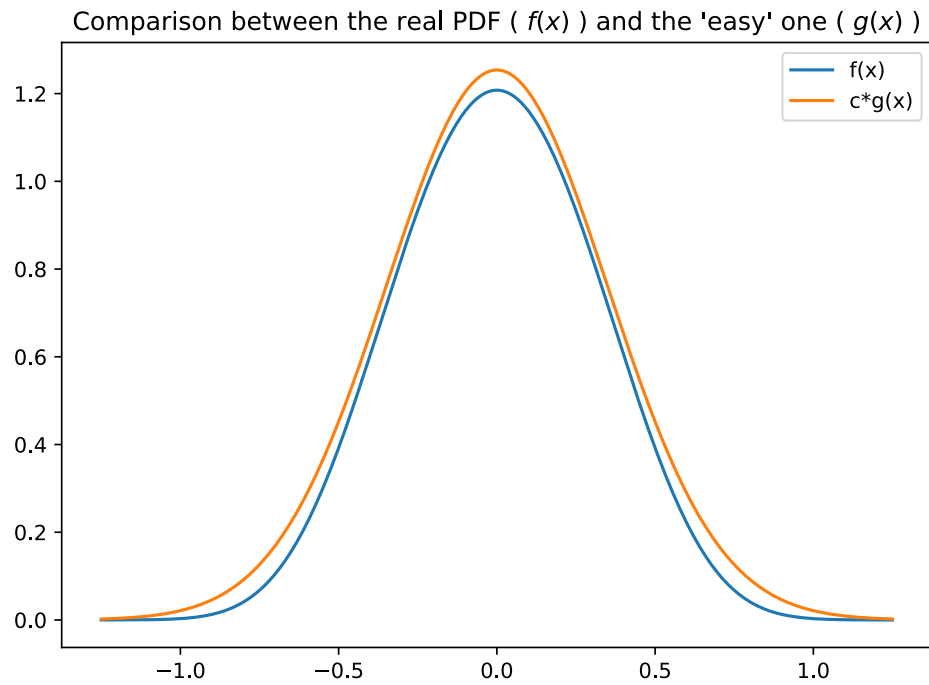


Figure 2.1

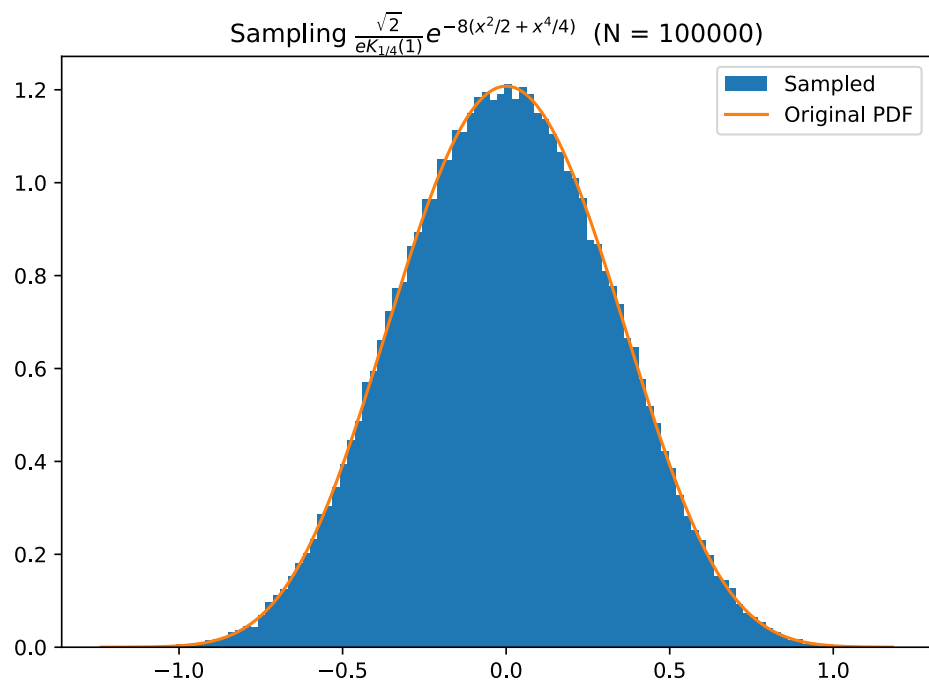


Figure 2.2

Exercises Lecture 3

3.1 Importance Sampling

Estimate the integral

$$\int_0^{\frac{\pi}{2}} \sin(x) dx \quad (3.1)$$

by

- The crude Monte Carlo method
- The importance sampling technique using a $g(x)$ proportional to $a + bx^2$

Determine the optimal values of the parameters a and b to generate samples according to $g(x)$ and establish the number of iterations N needed to get an accuracy of %1. Compare N with the number of iterations required to obtain the same accuracy for the crude Monte Carlo method.

3.1.1 Solution

The crude MC was easy to implement. Instead, finding the best a and b for the importance sampling was a bit more time consuming. Finding something similar to $\sin(x)$ with a quadratic term was a bit unexpected (especially without the linear term), in fact last year there was a similar exercise but with $\cos(x)$ instead of $\sin(x)$. In any case, the first idea was to use the Taylor series of the sine to find the best x_0 in the given interval:

$$T_{x_0}(x) = \sin(x_0) + \cos(x_0)(x - x_0) - \frac{\sin(x_0)}{2}(x^2 + x_0^2 - 2x_0x) \quad (3.2)$$

$$= \left(\sin(x_0) + x_0 \cos(x_0) - \frac{x_0^2 \sin(x_0)}{2} \right) + x (\cos(x_0) + x_0 \sin(x_0)) + x^2 \left(-\frac{\sin(x_0)x^2}{2} \right) \quad (3.3)$$

We want the constant term to be a and the quadratic term to be b . Since we don't want the linear term, we must impose

$$\cos(x_0) + x_0 \sin(x_0) = 0 \implies \tan(x_0) = -\frac{1}{x_0} \quad (3.4)$$

Sadly there are no solution for $x_0 \in [0, \pi/2]$. So we decided to find instead $g(x)$ in the form of $a + bx + cx^2$. Since $g(x)$ must be a PDF, $g(x) \geq 0$. Let's try to see what happens by requiring $g(0) \geq 0$:

$$g(0) = a = \sin(x_0) + x_0 \cos(x_0) - \frac{x_0^2 \sin(x_0)}{2} \stackrel{!}{\geq} 0 \implies \quad (3.5)$$

$$\sin(x_0) \left(1 - \frac{x_0^2}{2} \right) \geq x_0 \cos(x_0) \implies \tan(x_0) \geq \frac{2x_0}{2 - x_0^2} \quad (3.6)$$

which sadly has only one solution in the given interval: $x_0 = 0$. This implies $a = c = 0$ and $b = 1$: the only Taylor's series which has $g(0) \geq 0$ in $[0, \pi/2]$ is $x_0 = 0$ which is the well know $\sin(x) \approx x + \dots$

So we are now back to the starting point. Let's use $g(x) = a + bx^2$ now, and let's find a such that $g(x)$ is normalized:

$$\int_0^{\pi/2} (a + bx^2) dx \stackrel{!}{=} 1 \implies a = \frac{2}{\pi} \left(1 - \frac{b\pi^3}{24} \right) \quad (3.7)$$

Now let's require that $g(0) \geq 0$ and $g(\pi/2) \geq 0$:

$$g(0) \geq 0 \implies b \leq \frac{24}{\pi^3} \approx 0.774 \quad (3.8)$$

$$g(\pi/2) \geq 0 \implies b \geq -\frac{12}{\pi^3} \approx -0.387 \quad (3.9)$$

(it can be shown that $g(x)$ does not change sign in the interval given, so if the extremes are ≥ 0 then all $g(x)$ is). This gives us a range for the parameter b . We would like to find the b which minimize:

$$\int_0^{\pi/2} |\sin(x) - a(b) - bx^2| dx \quad (3.10)$$

but this seems a bit too much given that we want to calculate $\int_0^{\pi/2} \sin(x) dx$. So by using a graphic calculator we decided to use $b = 0.45$ because it seems to approximate better the sine for the importance sampling. (In figure 3.1 we put both curves for comparison).

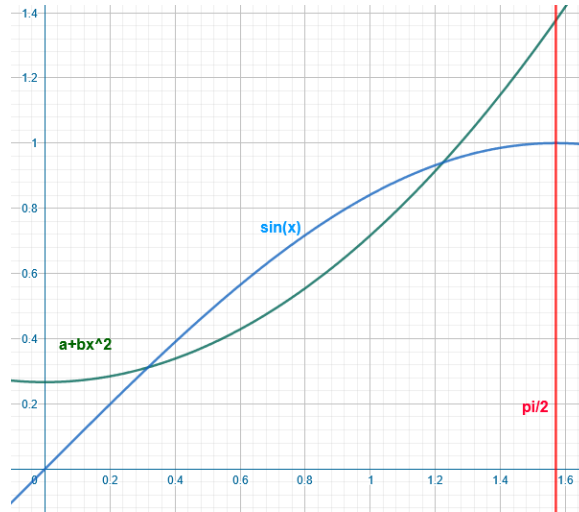


Figure 3.1

We then proceeded to find the iteration at which the deviation is $\leq 1\%$. The deviation is the absolute difference between the true value of 1 and the calculated value. We ran the experiment 1000 times to get the average iteration. We also added $g(x) = \frac{384}{48\pi^2 - \pi^4} \left(x - \frac{x^3}{6} \right)$, the third degree Taylor expansion in $x = 0$ normalized.

Average iteration Crude MC: 18.68

Average iteration Importance Sampling ($a+bx^2$): 11.86

Average iteration Importance Sampling (Third degree Taylor): 2.51

If we iterate for 10M:

Crude MC estimate: 1.00016334, err: 0.00016334

Importance Sampling estimate (a+bx^2): 1.00006674, err: 6.67401449e-05

Importance Sampling estimate (Third degree Taylor): 0.99999275, err: 7.24293925e-06

3.2 Importance Sampling

Let us compute the average

$$\langle f \rangle_\rho = \int_{\mathbb{R}} f(x) \rho(x) dx \quad (3.11)$$

where $\rho \in \mathcal{N}(0, 1)$ and

$$f(x) = e^{-(x-3)^2/2} + e^{-(x-6)^2/2} \quad (3.12)$$

- Show that $\langle f \rangle_\rho$ can be computed in closed form and derive its value.
- Construct a regular Monte Carlo approximation based on a normal distribution $\mathcal{N}(0, 1)$ by sampling $N = 10^3$ points and estimate the corresponding error.
- Compare the above result with the one obtained using the importance sampling approximation based on the uniform distribution $g \in \mathcal{U}[-8, -1]$. Use again $N = 10^3$ samples.

3.2.1 Solution

First, let's calculate $f(x)\rho(x)$:

$$f(x)\rho(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}} \left(e^{-(x-3)^2/2} + e^{-(x-6)^2/2} \right) = \frac{1}{\sqrt{2\pi}} \left(e^{-\frac{1}{2}(2x^2+9-6x)} + e^{-\frac{1}{2}(2x^2+36-12x)} \right) \quad (3.13)$$

$$= \frac{e^{-9/4}}{\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(x-3/2)^2}{1/2}} + \frac{e^{-9}}{\sqrt{2\pi}} e^{-\frac{1}{2} \frac{(x-3)^2}{1/2}} \quad (3.14)$$

so basically it's a gaussian with $\mu = 3/2$ and $\sigma^2 = 1/2$ since $e^{-9/4} \gg e^{-9}$. Then $\langle f \rangle_\rho$ will be:

$$\langle f \rangle_\rho = \int_{\mathbb{R}} f(x)\rho(x) dx = \frac{e^{-9/4}}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-\frac{1}{2} \frac{(x-3/2)^2}{1/2}} + \frac{e^{-9}}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-\frac{1}{2} \frac{(x-3)^2}{1/2}} \quad (3.15)$$

$$= \frac{e^{-9/4}}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(x-3/2)^2} + \frac{e^{-9}}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-(x-3)^2} \quad (3.16)$$

$$= \frac{e^{-9/4}}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-y^2} dy + \frac{e^{-9}}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-y^2} dy = \frac{e^{-9/4}}{\sqrt{2}} + \frac{e^{-9}}{\sqrt{2}} \quad (3.17)$$

We now use the regular Monte Carlo approximation sampling to approximate $\langle f \rangle_\rho$ (with $\rho(x) = \mathcal{N}(0, 1)$):

$$\langle f \rangle_\rho \approx \frac{1}{N} \sum_{i=1}^N f(x_i) \quad \text{with } x_i \sim \mathcal{N}(0, 1) \quad (3.18)$$

Instead, for the importance sampling, we have $\rho = \mathcal{N}(0, 1)$ and $g(x) = \mathcal{U}[-8, -1]$:

$$\langle f \rangle_\rho = \int_{\mathbb{R}} f(x)\rho(x) dx = \int_{\mathbb{R}} f(x) \frac{\rho(x)}{g(x)} g(x) dx = \int_{\mathbb{R}} F(x) g(x) dx = \quad (3.19)$$

$$= \langle F \rangle_g \approx \frac{1}{N} \sum_{i=1}^N f(x_i) \frac{p(x_i)}{g(x_i)} \quad \text{with } x_i \sim \mathcal{U}[-8, -1] \quad (3.20)$$

Note that the domain should be $\mathbb{R}$ but we are limited to $[-8, -1]$ so we are actually calculating $\int_{-8}^{-1} f(x)\rho(x)dx$.

Estim. value with regular MC: 0.074478, dev: 0.000138 (true value: 0.074616)

Estim. value with importance sampling: 1.484e-05, dev: 3.263e-07 (true value: 1.51648e-05)

The two results are not comparable because when we use the regular MC with $\rho = \mathcal{N}(0, 1)$ even if is not centered around $3/2$ its tails can reach every point (before vanishing) of $\rho(x)f(x)$. Instead, in the case of importance sampling, by using $g(x) = \mathcal{U}[-8, -1]$ we restrict ourselves to a specific interval, and so the results show different things.

Exercises Lecture 4

4.1 Markov chains

Draw the digraphs and classify the states of the Markov chains defined by the following transition matrices:

$$(A) \quad P = \begin{pmatrix} 0 & 0 & 0.5 & 0.5 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad (B) \quad P = \begin{pmatrix} 0.3 & 0.4 & 0 & 0 & 0.3 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.6 & 0.4 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

4.1.1 Solution

The digraphs are:

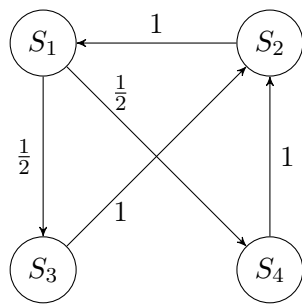


Figure 4.1: (A)

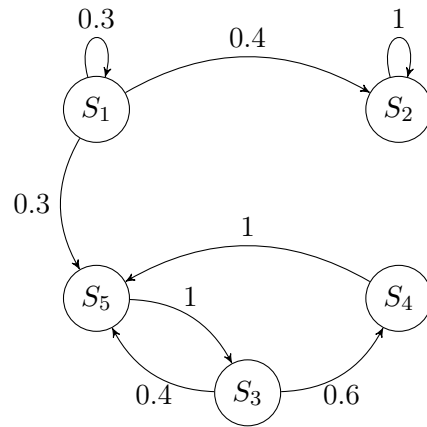


Figure 4.2: (B)

For (A) all states are *recurrent*, since if I start on anyone of them I can always return. In addition, all states are *periodic* with a period of 3.

Instead for transition matrix (B), all states are *recurrent* except for the state S_1 that is transient since there is a possibility to be able to never return to that state (for example going from S_1 to S_2 or to S_5). Also, all states here are *aperiodic* (for example I can return to $S_{3/4/5}$ after any $M \geq 3$) except for S_4 since I can return to that state only after $2n + 1$ steps (with $n \geq 1$).

4.2 Markov chains

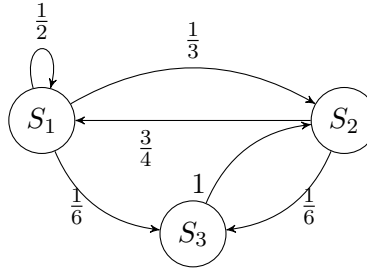
Consider the Markov chain with transition matrix:

$$P = \begin{pmatrix} 1/2 & 1/3 & 1/6 \\ 3/4 & 0 & 1/4 \\ 0 & 1 & 0 \end{pmatrix}$$

- (a) Show that the Markov chain is irreducible and aperiodic.
- (b) Suppose the process starts in state 1; find the probability that it is in state 3 after two steps.
- (c) Find the matrix which is the limit of P^n as $n \rightarrow \infty$.

4.2.1 Solution

(a) By looking at the graph we can easily see that the Markov chain is irreducible since every node can be reached from every node. Since S_1 is aperiodic because of the self-path, then all other states are aperiodic since the chain is irreducible (since there is a path to every node, we can always pass by S_1 and do as many loop as we want thus making every node aperiodic).



(b) By calculating $x^T P^2$ with $x^T = (1, 0, 0)$ we can find the probability to be in S_3 after two steps starting in S_1 :

$$x^T P^2 = (1, 0, 0) \begin{pmatrix} 1/2 & 1/3 & 1/6 \\ 3/4 & 0 & 1/4 \\ 0 & 1 & 0 \end{pmatrix}^2 = (1/3, 1/2, 1/6)$$

The probability to be in S_3 is $1/6$. We can also do it by hand since it's easy in this case:

$$P(S_1^1; S_1^2; S_3^3) = \frac{1}{12} \quad P(S_1^1; S_2^2; S_3^3) = \frac{1}{12} \quad P(S_1^1; S_3^2; S_3^3) = 0 \implies P(S_1^1; S_3^3) = \frac{1}{12} + \frac{1}{12} \quad (4.1)$$

(c) To find P^n it's better to diagonalize $P = QDQ^{-1}$ because then:

$$P^n = (QDQ^{-1})^n = QDQ^{-1} \cdot QDQ^{-1} \dots QDQ^{-1} = QD^nQ^{-1} \quad (4.2)$$

So let's diagonalize P in QDQ^{-1} :

$$Q = \frac{1}{6} \begin{pmatrix} 0 & -2 & 6 \\ -3 & 0 & 6 \\ 6 & 6 & 6 \end{pmatrix} \quad D = \frac{1}{2} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad Q^{-1} = \frac{1}{6} \begin{pmatrix} 6 & -8 & 2 \\ -9 & 6 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$D^n = \begin{pmatrix} -\frac{1}{2^n} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad D^\infty = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

So then

$$P^\infty = QD^\infty Q^{-1} = \frac{3}{16} \begin{pmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{pmatrix}$$

We can test that

$$PP^\infty = \frac{1}{12} \begin{pmatrix} 6 & 4 & 2 \\ 9 & 0 & 3 \\ 0 & 12 & 0 \end{pmatrix} \frac{3}{16} \begin{pmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{pmatrix} = \frac{3}{16} \begin{pmatrix} 3 & 2 & 1 \\ 3 & 2 & 1 \\ 3 & 2 & 1 \end{pmatrix}$$

4.3 Markov chains

Two boxes, A and B, contain 2 white and 3 red balls, respectively. At each time, one extracts one ball from each box and puts it back in the opposite box (1 event). Let us call a_i the state corresponding to the number i of red balls in box A. Determine the occurrence probability of state a_2 after 3 events and in the limit of many events.

4.3.1 Solution

We assume that there is a typo and a_i (and not a_1) is the state corresponding to the number i of red balls in box A. So a_2 is the state when there are 2 red balls in A. We start by calculating the transition probability (where “ $\circ$; $\bullet$ ” means “I drew a white ball from A and a red ball from B”):

$$\text{For state } a_0: \quad P(a_0 \rightarrow a_1) = P(\circ; \bullet | a_0) = 1 \quad P(a_0 \rightarrow a_{i \neq 1}) = 0 \quad (4.3)$$

$$\text{For state } a_1: \quad P(a_1 \rightarrow a_0) = P(\bullet; \circ | a_1) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} \quad (4.4)$$

$$P(a_1 \rightarrow a_1) = P(\bullet; \bullet | a_1) + P(\circ; \circ | a_1) = \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{2} \quad (4.5)$$

$$P(a_1 \rightarrow a_2) = P(\circ; \bullet | a_1) = \frac{1}{3} \quad (4.6)$$

$$\text{For state } a_2: \quad P(a_2 \rightarrow a_0) = 0 \quad (4.7)$$

$$P(a_2 \rightarrow a_1) = P(\bullet; \circ | a_2) = \frac{2}{3} \quad (4.8)$$

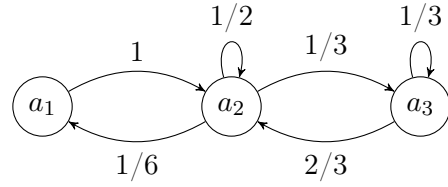
$$P(a_2 \rightarrow a_2) = P(\bullet; \bullet | a_2) = \frac{1}{3} \quad (4.9)$$

From which we can get the transition matrix and the directed graph:

To get the probability to go from state a_0 to state a_2 in three steps, we calculate $x^T P^3$ where $x^T = (1, 0, 0)$:

$$P^3 = \begin{pmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix}^3 = \frac{1}{216} \begin{pmatrix} 18 & 138 & 60 \\ 23 & 127 & 66 \\ 20 & 132 & 64 \end{pmatrix} \Rightarrow (1, 0, 0)P^3 = \left(\frac{3}{36}, \frac{23}{36}, \frac{10}{36} \right)$$

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 0 & 6 & 0 \\ 1 & 3 & 2 \\ 0 & 4 & 2 \end{pmatrix}$$



From which we get the probability of $\frac{10}{36}$. Instead if we want the probability in the limit of many events we need to diagonalize to raise the matrix to the power of n as before:

$$P = QDQ^{-1} \quad \text{where} \quad Q = \frac{1}{4} \begin{pmatrix} 12 & -6 & 4 \\ 4 & -1 & 4 \\ 4 & 4 & 4 \end{pmatrix} \quad D = \frac{1}{6} \begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{pmatrix} \quad Q^{-1} = \frac{1}{30} \begin{pmatrix} 5 & -10 & 5 \\ -8 & -8 & 16 \\ 3 & 18 & 9 \end{pmatrix}$$

from which

$$D^n = \begin{pmatrix} -\frac{1}{3^n} & 0 & 0 \\ 0 & -\frac{1}{6^n} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad D^\infty = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

So then

$$P^\infty = QD^\infty Q^{-1} = \frac{1}{10} \begin{pmatrix} 1 & 6 & 3 \\ 1 & 6 & 3 \\ 1 & 6 & 3 \end{pmatrix}$$

So if we start in the state a_0 , the probability of being in state a_2 in the limit of many events is:

$$(1, 0, 0)P^\infty = (1/10, 6/10, 3/10) \implies P(a_2^\infty | a_0^0) = 3/10$$

Exercises Lecture 5

5.1 MC simulation of a 2D Ising model by the Metropolis algorithm

Program Write a program that simulates a 2D Ising model with periodic boundary conditions by using the Metropolis acceptance matrix a_{ij} and the matrix Γ_{ij} based on the local spin-flip (Glauber) proposed move.

Simulation By assuming $k_B = 1$ and $J = 1$ simulate the 2D Ising model for different values of temperatures (at least 3, one below, one above, and one close to the critical temperature $T_c = \frac{2}{\ln(1+\sqrt{2})}$) and 3 different values of L (for instance 25, 50 and 100).

Equilibration time, averages and fluctuations After having determined the equilibrium time and disregarding the samples for $t < \tau_{eq}$ estimate the ensemble averages of the magnetization per spin, the energy per spin, and the corresponding fluctuations (specific heat and magnetic susceptibility).

Integrated correlation time and critical slowing-down Estimate the autocorrelation time of the magnetization and the energy for the MC simulations proposed above and estimate the errors accordingly.

5.1.1 Solution

The energy of this 2D Ising model given a configuration of the spins $\mathcal{C}$ is:

$$H(\mathcal{C}) = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j \quad \text{with } \sigma_i = \pm 1 \quad \text{and} \quad \langle i,j \rangle \quad \text{it's a sum over n.n.} \quad (5.1)$$

At every step of the MC, all spins are selected in order and for each we calculate the energy difference if that spin is flipped as:

$$\Delta E = 2J\sigma_i \sum_{\langle j \rangle_i} \sigma_j \quad \text{with } \langle j \rangle_i \text{ the 4 neighbors of } \sigma_i \quad (5.2)$$

If $\Delta E < 0$ or $\exp\{(-\Delta E)\beta\}$ is bigger than a random number between $[0, 1]$ then we accept the move and flip the spin (this is the Metropolis algorithm).

For each MC steps, we calculate our observables: the energy and magnetization per site. After the system reaches equilibrium, the averages of the observable are calculated, from which we can calculate the specific heat and the magnetic susceptibility.

$$\langle E_{site} \rangle = \frac{1}{t_{max} - \tau_{eq}} \frac{1}{L^2} \sum_{t=\tau_{eq}}^{t_{max}} H_t \quad \langle m \rangle = \frac{1}{t_{max} - \tau_{eq}} \frac{1}{L^2} \sum_{t=\tau_{eq}}^{t_{max}} M_t \quad C_V = \frac{\text{Var}[E_{site}]}{k_B T^2} \quad \chi_T = \frac{\text{Var}[m]}{k_B T} \quad (5.3)$$

We also calculated:

- τ_{eq} the time the simulation reached the equilibrium

- τ_O^{int} the integrated autocorrelation time which is:

$$\int_0^{t_{max}} \frac{C_O(t)}{C_O(0)} dt = \int_0^{t_{max}} e^{-t/\tau} dt = \tau_O^{int} = \sum_{t=0}^{t_{max}} \frac{C_O(t)}{C_O(0)} \Delta t \quad (5.4)$$

which tells us how distant we should sample O to get uncorrelated samples ($C_O(t)$ is the autocorrelation function).

- $S_{t_{max}}^2$ which tells us the variance of the mean of the observable O and it is calculated as:

$$S_O^2 = \frac{1 + 2\tau/\Delta t}{(t_{max} - \tau_{eq} - 1)} \sum_{t=\tau_{eq}}^{t_{max}} (O_t - \bar{O}_{t_{max}})^2. \quad (5.5)$$

A C++ program calculates the evolution of the spins on the lattice and writes them to a binary file at every step. Then a Python script reads it and analyzes the content.

We run the program with 3 different lattice dimensions (side): 50, 100 and 200, and for 3 different values of temperature: $0.5T_c$, $0.97T_c$ and $1.2T_c$.

The results are shown in table 5.1.

Results									
	$T = 0.5 T_c$			$T = 0.97 T_c$			$T = 1.2 T_c$		
	$L = 50$	$L = 100$	$L = 200$	$L = 50$	$L = 100$	$L = 200$	$L = 50$	$L = 100$	$L = 200$
$\langle E_{site} \rangle$	-1.993	-1.992	-1.992	-1.541	-1.546	-1.546	-0.95	-0.9501	-0.9501
$\langle m \rangle$	-0.9981	-0.9981	-0.998	0.7735	-0.7845	-0.7863	$-1.37 \cdot 10^{-3}$	$2.15 \cdot 10^{-3}$	$2.37 \cdot 10^{-4}$
N_{steps}	2000	2000	2000	16000	16000	16000	8000	8000	8000
C_V	0.04523	0.04791	0.05224	1.472	1.439	1.32	0.6411	0.5773	0.5829
χ_T	0.00363	0.003803	0.004164	7.104	4.44	3.626	8.195	7.438	8.45
τ_{eq}	100	100	200	500	4600	9600	100	4200	800
τ_E^{int}	1.01	1.74	0.967	30.6	29.9	14.5	29.7	5.76	8.33
τ_m^{int}	0.967	1.03	0.857	27.6	75.6	45.9	15.6	8.04	6.95
S_E^2	$7.02 \cdot 10^{-5}$	$2.76 \cdot 10^{-5}$	$4.93 \cdot 10^{-6}$	$1.78 \cdot 10^{-1}$	$4.25 \cdot 10^{-2}$	$4.8 \cdot 10^{-3}$	$1.148 \cdot 10^{-1}$	$5.36 \cdot 10^{-3}$	$1.91 \cdot 10^{-3}$
S_m^2	$4.85 \cdot 10^{-6}$	$1.32 \cdot 10^{-6}$	$3.21 \cdot 10^{-7}$	$3.51 \cdot 10^{-1}$	$1.49 \cdot 10^{-1}$	$1.851 \cdot 10^{-2}$	$2.87 \cdot 10^{-1}$	$3.46 \cdot 10^{-2}$	$8.57 \cdot 10^{-3}$

Table 5.1: Results for Ising 2D

We also show some images for $L = 200$ of the E_{site} and m for the three temperatures: figure 5.1. On github are available also the animation from $t = 0$ to $t \approx \tau_{eq}$: [here](#).

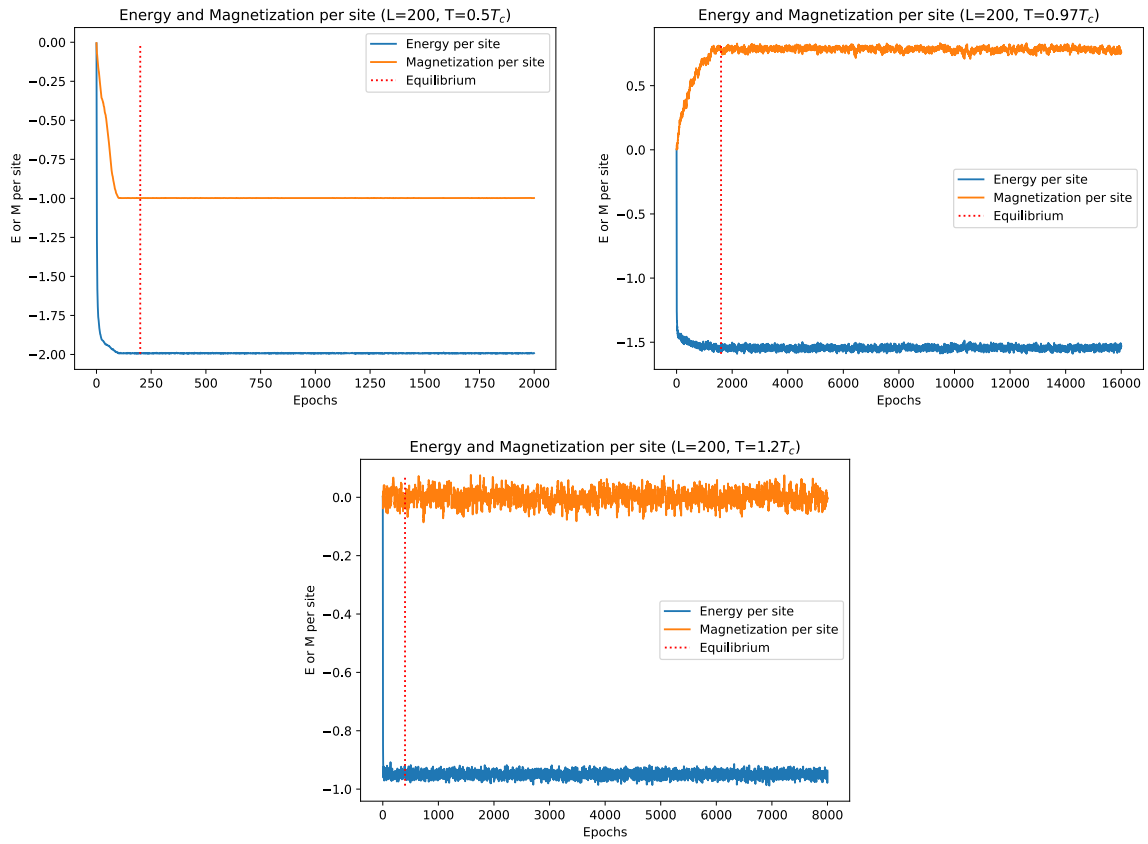


Figure 5.1: Energy and magnetization for three systems with $L = 200$ and different temperatures. The red line is when the equilibrium was reached (for both quantities).

Exercises Lecture 6

6.1 Exercise on multiple Markov chains (MMC)

Simulate the absorption of a grafted polymer to an attractive hard wall with a Monte Carlo using the MMC method. The model for the polymer is composed of $N + 1$ monomers in the two-dimensional plane, with coordinates $\vec{r}_i = (x_i, y_i)$ for $0 \leq i \leq N$.

Geometry

- Grafting is imposed by fixing $(x_0, y_0) = (0, 0)$.
- Self-avoidance between any monomers i and j is imposed by requiring $|\vec{r}_i - \vec{r}_j| > c$ with $c = 1$
- The chain connectivity is achieved by requiring each bond length $l_i = |\vec{r}_i - \vec{r}_{i+1}|$ to fluctuate within the interval $[l^{\min}, l^{\max}]$ with say $l^{\min} = c$ and $l^{\max} = 1.3c$.
- The hard wall constraint imposes $y_i \geq 0$ for each monomer i .

Thermodynamics

- The attraction between the wall on the monomers derives from an energetic contribution $-\epsilon = -1$ to each monomer with $y_i < h$ say with $h = 1/2$. We may neglect the trivial constant contribution from the grafted monomer $i = 0$. Thus, the total energy of a configuration $\mathcal{C}$ is $E(\mathcal{C}) = -\epsilon n$ where n is the number of monomers in $\mathcal{C}$ with $y_i < h$.
- Each polymer is in a canonical ensemble at its own inverse temperature $\beta \geq 0$ (β 's will be different; see the MMC scheme below).

Monte Carlo moves

- Local shifts: choose a displacement $\vec{\delta}$ from a symmetric distribution (e.g., a Gaussian around zero or an appropriate uniform distribution) and try to displace a randomly picked monomer $\vec{r}_i \rightarrow \vec{r}_i + \vec{\delta}$. If the geometric constraints are not satisfied, reject the move; otherwise, accept the new configuration $\mathcal{C}'$ according to the Metropolis rule at β .
- Pivot: choose a monomer p at random in the range $0 \leq p \leq N - 1$ and rotate the monomers with $i > p$ as a rigid body with respect $\vec{r}_p$ of an angle drawn randomly from an interval $[-\theta^*, \theta^*]$. Again, after checking if the geometry is fine, accept the move to $\mathcal{C}'$ according to the Metropolis rule. Note that the difference in energy $E(\mathcal{C}') - E(\mathcal{C})$ depends only on the changes for $i > p$.

MMC

- Choose a reasonable set of K inverse temperatures $\{\beta_k\}$, in python or C language indexed from $k = 0$ to $k = K - 1$ at larger β' s, the polymer should be mostly absorbed; at small β it should float freely; at a critical β_c , it should display the largest fluctuations in energy and other geometrical indicators.
- We do not want to copy configurations in a swap because it takes $2 \times N \times 2$ operations. It is convenient to track which configuration belongs to a given β_k using an array of indices I . Initially $I = (I_0 = 0, I_1 = 1, \dots, I_{K-1} = K - 1)$. Then, each time there is a swap at $k, k + 1$, only indices swap: $(I_0, I_1, \dots, I_k = A, I_{k+1} = B, \dots, I_{K-1}) \rightarrow (I_0, I_1, \dots, I_k = B, I_{k+1} = A, \dots, I_{K-1})$

In this scheme, the moves at β_k act on its current configuration, stored at position I_k . This is the first index in a multidimensional matrix (the second is the monomer index i , where its (x_i, y_i) are found). In summary, at β_k , we have $r(I_k, i, 0)$ for the x_i of the configuration and $r(I_k, i, 1)$ for y_i .

- As a typical recipe, for every β_k , a block of moves contains one attempted pivot and N attempted local shifts. Furthermore, one swap between configurations at neighboring β_k and β_{k+1} (with randomly picked $k \in [0, K - 2]$) should be attempted at the end of the K blocks. This recipe is by no means rigid; it is just an example.

Between samplings, B blocks of moves should be performed to decorrelate data sufficiently.

- To monitor the state of the system, evaluate:
 - The average energy $U(\beta) = \langle E \rangle_\beta$ and its variance $VU(\beta) = \langle E^2 \rangle_\beta - \langle E \rangle_\beta^2$.
 - The mean squared end-to-end distance $R_e^2(\beta) = \langle |\vec{r}_N|^2 \rangle_\beta$, its variance $VR_e^2(\beta)$.
 - The mean height of the end, $Y_e(\beta) = \langle y_N \rangle_\beta$, and its variance $VY_e(\beta)$.

Questions

1. Show the energy histograms for all β_k in the same plot. Is there a good overlap between histograms of neighboring β_k ? What was the logic for settling down to a final list of β' s after your initial attempts?
2. Plot the listed quantities as a function of β for a given N or many N' s. Do the plots meet your expectations? Is there any signature of the critical temperature?
3. What did you find an appropriate distribution for the random displacements in the local shifts?
4. Why do we use a symmetric interval $[-\theta^*, \theta^*]$ in the pivot move? Should θ^* depend on β ?
5. Given the same B blocks of moves at all β_k how does the autocorrelation in data vary with β ?

6.1.1 Solution

To simulate the problem we have chosen

- $l^{min} = c = 1$, the diameter of the monomers and $l^{max} = 1.3c = 1.3$,
- $\theta^* = \pi/8$,
- N , the number of monomers excluding the first, equals to 25, 50, 75, 100, 150 and 200
- $K = 30$, the number of different polymers chains,
- $n_{shift} = 10$ the number of shifts for each pivot (the number of shifts per block),
- $n_{block} = 100$ the number of blocks between each sample,
- n_{iters} the number of samples (it changes depending of N)

To be clear, one block consists of updating all the K 's chains, after n_{block} a sample is taken from each chain. So, between two samples from the same chain, $n_{shift} \cdot n_{block}$ shifts and n_{block} pivots are attempted. The swap between two chains is attempted before sampling. The Metropolis rule, respectively for the local move and the chain swap, is

$$p_{CC'} = \min\left(1, e^{-\beta \Delta E}\right) \quad p_{CC'} = \min\left(1, e^{\Delta \beta \cdot \Delta E}\right) \quad (6.1)$$

The first thing was to plot the histograms of the energy for different sets of β 's without any chain swapping to have an idea of what is happening. The results are shown in figure 6.1 ($N = 100$ and $N_{iters} = 10000$).

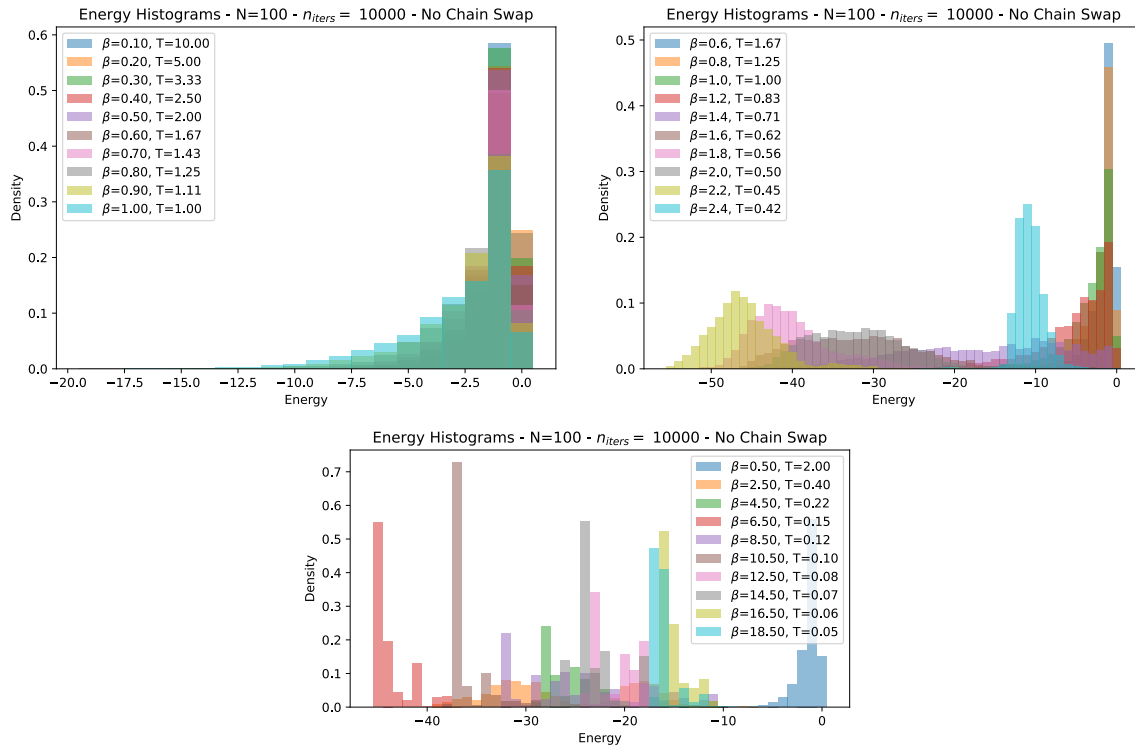


Figure 6.1: Histograms of the energies for different sets of β s without chain swapping. If the values are very low (high temperature, first image) they all display the same spectrum of energy. In the other two cases the histograms are more separated. Notice how the highest β s are not always associated with the lower energies. The images are vector images (they are zoom friendly).

The first image clearly tells us that those β 's are too low, so the temperature is very high and there is basically no absorbing behavior. Instead, in the other two images, the histograms are more distributed. It is interesting to note that the lowest temperatures are not always associated with the lowest energies, usually because the polymer is stuck at a local minimum of the energy. In fact, in figure 6.2 are shown all the shapes assumed by the polymer (with transparency) and the average position of its monomers; it got stuck in a local minimum and never recovered from it.

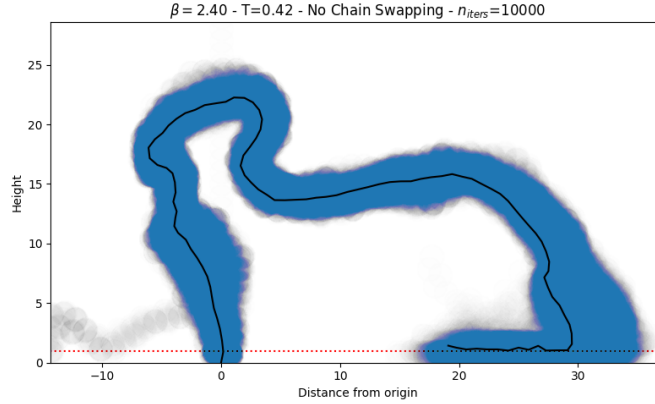


Figure 6.2: Heatmap of the position of the monomers for a low temperature with $N = 100$ for $\beta = 2.4$ as seen in the second image of figure 6.1. By comparing this with that figure (where the lower $\beta = 2.2$ easily reach $E = -45$), it is clear that this is stuck in a local minimum of the energy. The full flat position has less energy but to reach it requires passing by configurations at higher energy and at this low temperature is extremely unlikely.

If instead we add the possibility of chain swapping (excluding the sets of low betas), the results are more as expected, with lowest temperature chains associated with the lowest energies.

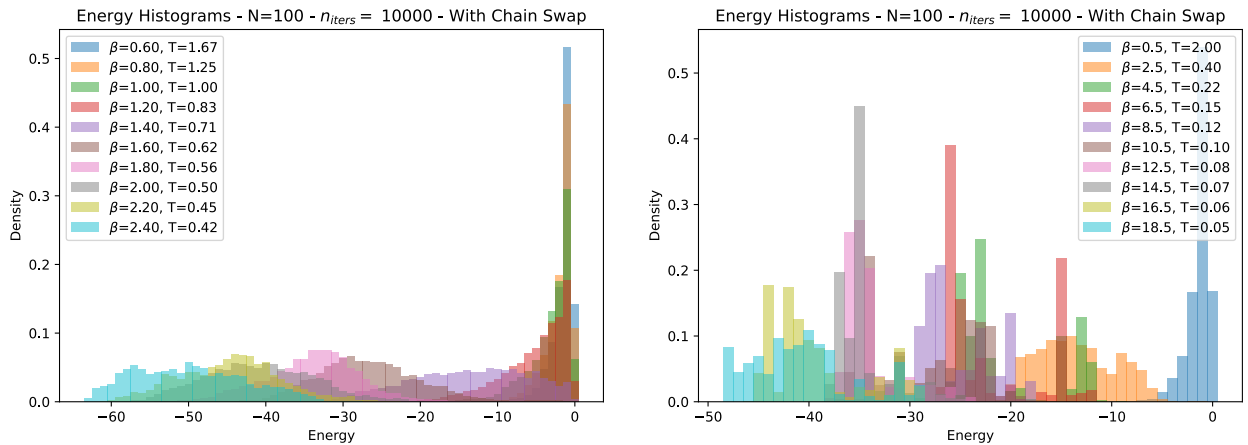


Figure 6.3: Histograms of the energies for different sets of β s with chain swapping. The sets are the second and third sets from the previous histograms (figure 6.1). Here we can see the trend of lower temperatures associated with higher energies. The images are vector images (they are zoom friendly).

We need to decide some ranges of β s. It has been suggested that the distribution of the energy of each β s should overlap between the neighbors for $\approx 20\%$. If the β 's are closer to each other, they tend to overlap more, so it would be natural to choose a spacing that accounts for that. We have preferred

to choose closer β s even if they overlap a lot because we would like to have a lot of point in a particular range of temperatures. We decided that is better to swap more than swap less, and also it is worth notice that only one chain is swapped at random before sampling all the chains. Bigger K means that chains are in general swapped less frequently, so also this should be take in account when deciding the percentage of swapping.

The range we have chosen is $\beta_i = 2/3 + 0.12 \cdot i$ with $i \in [0, 29]$, so $\beta_{29} \approx 4.15$, because we saw a signature of a phase transition in that range. The initial configuration for every polymer was the one with every monomer stacked one on top of the other at $x = 0$. Then we proceeded to calculate the mean values for the observables and put them in figure 6.4. In figure are shown the lines for different values of N . The second row shows the same observables but the one on the right is divided by N .

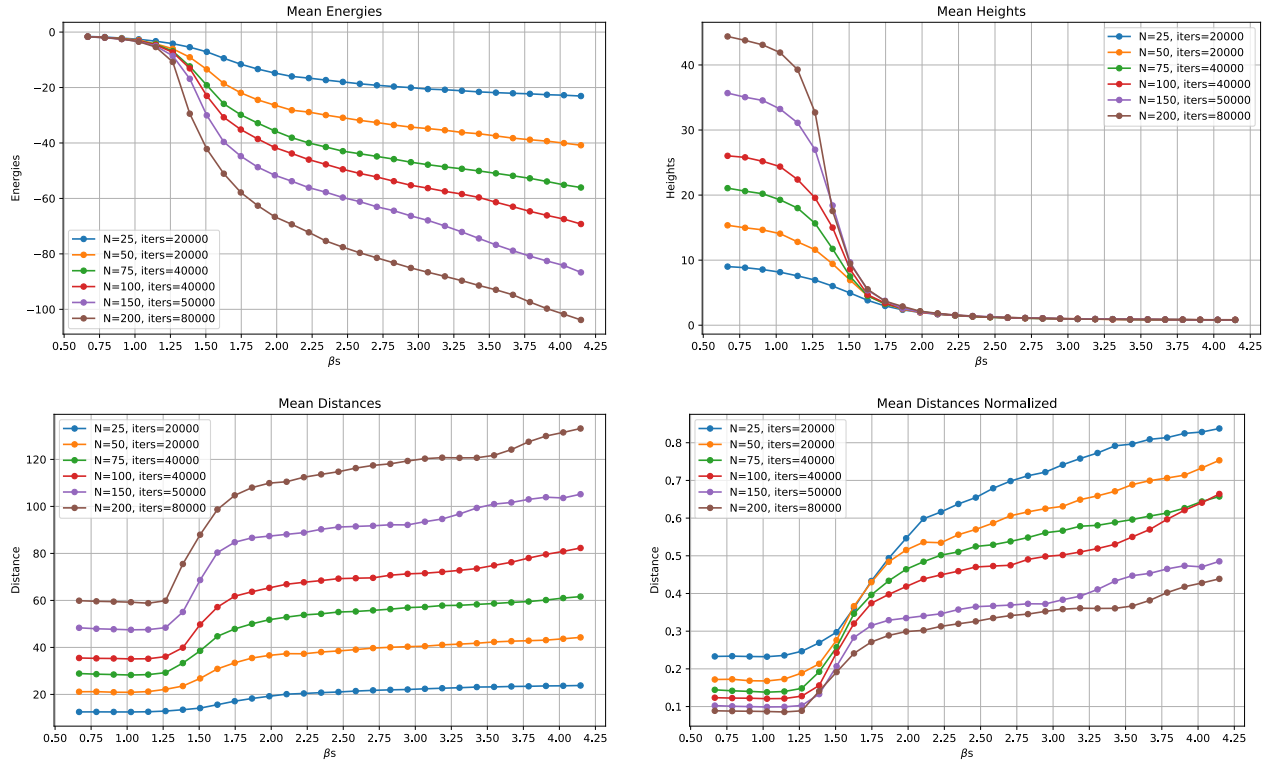


Figure 6.4: In the first row are shown the mean energies with the mean heights of the last monomer. In the second row are shown the mean distances of the chains; the one on the right is the same plot but the distances are divided by N . In theory the maximum distance could be $N \cdot l^{max} = 1.3N$ but there is no incentive in being longer (one could say that it's easier to get less entangled, that's true), only the energy matters and so dividing by N is enough! It is very clear from the images on the right that there is a critical point around $\beta = 1.5$.

We prefer showing the distance and not the squared distance, specifically the norm of the vector starting in $(0, 0)$ and ending in the center of the last monomer. We did this because the plot on the bottom left can be visualized better; in both cases, the trend stays. There is a clear mark of a critical point, especially in the bottom right image. It is not exactly an order parameter because it is not 0 before the critical point, but there is potential to find a better one by shifting it by the average distance when there is no absorbing wall.

About the distribution of the local shifts, we have chosen to go simple by using an uniform distribution $\mathcal{N}(-\frac{3c}{4}, \frac{3c}{4})$. Going more than $0.75c$ was just wasteful because it is very close to the maximum stretch possible. Given this distribution, if we have 3 monomers aligned horizontally, the one in the

middle can stretch vertically up to $y = 0.75c$ which make the distances between the others equals to $\frac{5c}{4} = 1.25c$ that is very close to the maximum allowed of $1.3c$. Obviously this distribution could also be gaussian with a very low deviation, to make small but likely legal moves, but we have chosen simplicity given all the degrees of freedom of the problem.

Instead, about the θ^* we have chosen to draw it uniformly from the interval $[-\pi/8, \pi/8]$, there is not a specific reason but allowing bigger angles would have caused a lot of illegal moves.

It makes sense to be symmetric because there is no reason to prefer one side or the other given the symmetry of the problem. Unless, we consider some potential between the monomers which make them attractive and thus creating a bias

As suggested would make sense to make $\theta^* = \theta^*(\beta)$ because higher temperatures mean higher fluctuations in energy and so bigger movements (bigger angles). I would also personally suggest to make θ^* dependent on the index of the monomers: a pivot closer to the origin means moving all the other monomers which should costs more and so be less probable. On the opposite, bigger rotations should be more likely near the end of the polymer.

Instead, I would add the move “global shift”: select a random monomers p , then select an uniform value between $[-0.1c, 0.25c]$. This value should be added to the actual distance between the p and the $p + 1$ monomer (keeping the direction the same), thus rigidly shifting all the monomers which come after.

By doing some testes, the autocorrelation tends to increase as β increases.

An animation can be seen [here on Github](#).

Exercises Lecture 7

7.1 Gillespie algorithm

Implement the Gillespie algorithm to simulate the Brusselator model, whose transition rates for a state $C = (X, Y)$ are

$$w_1 = a\Omega \quad \text{for } (X, Y) \rightarrow (X + 1, Y) \quad (7.1)$$

$$w_2 = X \quad \text{for } (X, Y) \rightarrow (X - 1, Y) \quad (7.2)$$

$$w_3 = \frac{1}{\Omega^2} X(X - 1)Y \quad \text{for } (X, Y) \rightarrow (X + 1, Y - 1) \quad (7.3)$$

$$w_4 = bX \quad \text{for } (X, Y) \rightarrow (X - 1, Y + 1) \quad (7.4)$$

Run some simulations using $a = 2$, $b = 5$ and for different volume sizes: $\Omega = 10^2, 10^3, 10^4$. What can one note by varying Ω ?

7.1.1 Solution

We implemented the Brussellator with the Gillespie algorithm using $a = 2$, $b = 5$ and different values of Ω . The initial configuration was $X_0 = Y_0 = 2.5\Omega$. By increasing Ω we noticed that the τ (the time for any reaction to occurs next) was decreasing. This makes sense because of how the τ is calculated, but also because if the volume is bigger more reactions are expected, thus lowering the time between two reactions. We also noticed that by increasing the volume the oscillation period starts to decrease in the same amount of time.

The algorithm starts with calculating the rates of the reaction w_i with equation (7.1) to (7.4). Then we calculate τ of the next reaction as:

$$\tau = \frac{1}{\sum_i w_i} \ln \frac{1}{u} \quad \text{where } u \sim \mathcal{U}[0, 1] \quad (7.5)$$

Now that we know when the next reaction will occur, we need to know which one is it. We select the first k for which:

$$\sum_{j=0}^k a_j > u \sum_i a_i \quad \text{where } u \sim \mathcal{U}[0, 1] \quad (7.6)$$

The next reaction will be the k -th reaction chosen with probability $a_k / \sum_i a_i$. We then increment the time by t and update the current state based on the reaction chosen.

The results are shown in figure 7.1.

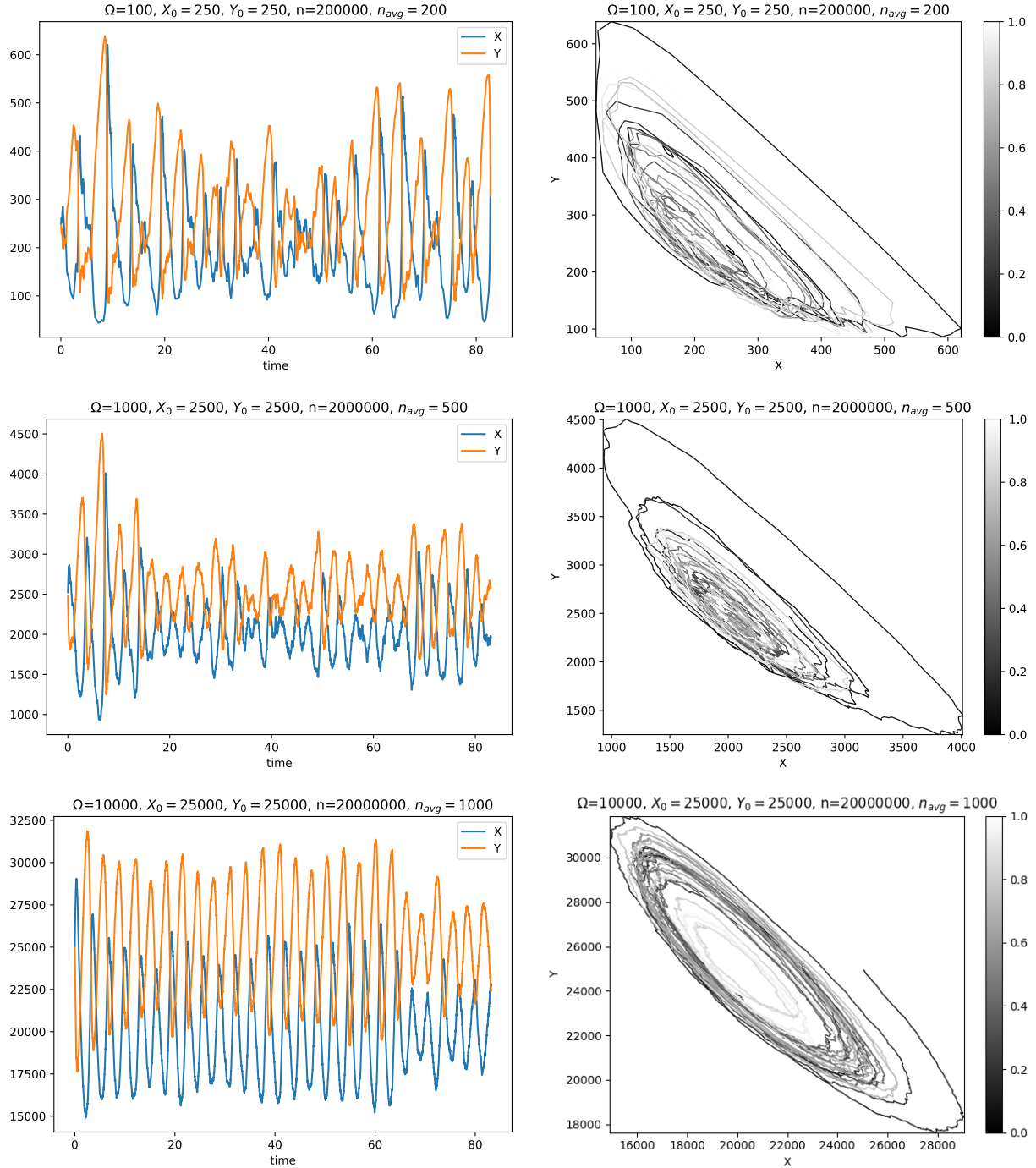


Figure 7.1: We increased the number of samples for different $\Omega = 10^2$, 10^3 and 10^4 , specifically $n_{iters} = 2000\Omega$. To avoid plotting too many points and to remove all the noise we decided to reduce the number of points with an average. Specifically every n_{avg} points the average is calculated and the result will be used instead of the points. The lines on the right are colored from black to light gray to show the path. (The last image is not a vector image because of too much points)

Exercises Lecture 8

8.1 Off-lattice Monte Carlo simulations - Reduced Units (pen & paper)

Typical sets of parameters for Argon and Krypton are $\sigma_{Ar} = 3.41\text{\AA}$, $\sigma_{Kr} = 3.38\text{\AA}$ for their typical size and $\varepsilon_{Ar}/k_B = 119.8\text{ K}$ and $\varepsilon_{Kr}/k_B = 164.0\text{ K}$.

- At the reduced temperature $T^* = 2$, what is the temperature of Argon in Kelvin?
- A typical value of the integration time step for MD is $\Delta t = 0.001\tau$. Convert it into SI units for Argon and Krypton.

8.1.1 Solution

Since $T = E/k_B$ then:

$$T^* = \frac{T}{T^*} = \frac{T}{\varepsilon/k_B} \implies T = \frac{\varepsilon}{k_B} T^* \quad (8.1)$$

Since we have $\varepsilon_{Ar}/k_B = 119.8$ and $T^* = 2$ then $T_{Ar} = 239.6\text{ K}$. For the Krypton instead $T_{Kr} = 328\text{ K}$.

About the reduced time τ , this can be obtained by using the other units: $\tau = \sigma\sqrt{\bar{m}/\varepsilon}$, but we still need the unit mass $\bar{m}$. We choose as the unit mass the atomic mass of the nuclei. So, we get $\bar{m}_{Ar} = 39.95u$ and $\bar{m} = 83.80u$ with $u = 1.661 \cdot 10^{-27}$. Putting these inside the formula for τ we get:

$$\tau_{Ar} = \sigma_{Ar} \sqrt{\bar{m}_{Ar}/\varepsilon_{Ar}} = 3.41 \cdot 10^{-10} \text{ m} \sqrt{\frac{39.95 \cdot 1.661 \cdot 10^{-27} \text{ kg}}{119.8 \text{ K} \cdot 1.381 \cdot 10^{-23} \text{ J/K}}} = 2.16 \cdot 10^{-12} \text{ s} \quad (8.2)$$

$$\tau_{Kr} = \sigma_{Kr} \sqrt{\bar{m}_{Kr}/\varepsilon_{Kr}} = 3.38 \cdot 10^{-10} \text{ m} \sqrt{\frac{83.80 \cdot 1.661 \cdot 10^{-27} \text{ kg}}{164.0 \text{ K} \cdot 1.381 \cdot 10^{-23} \text{ J/K}}} = 2.65 \cdot 10^{-12} \text{ s} \quad (8.3)$$

$$(8.4)$$

So in the end $\Delta t_{Ar} = 2.16 \cdot 10^{-15} \text{ s}$ and $\Delta t_{Kr} = 2.65 \cdot 10^{-15} \text{ s}$.

8.2 Off-lattice Monte Carlo simulations - Lennard-Jones

The Lennard-Jones interaction is a paradigmatic interaction potential, used in different contexts. It reads

$$V_{LJ}(r) = \begin{cases} 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] & \text{for } r < \sigma_{cut} \\ 0 & \text{for } r \geq \sigma_{cut} \end{cases} \quad (8.5)$$

where $\sigma = 1$, $\epsilon = 1$ are the units of length and energy and σ_{cut} is the so-called cut-off length and is set, for this exercise, to half the box size. Beyond this cut-off, we introduce two tail corrections: one for the energy (per particle)

$$u^{tail} = \frac{8}{3}\pi\rho \left[\frac{1}{3} \left(\frac{\sigma}{\sigma_{cut}} \right)^9 - \left(\frac{\sigma}{\sigma_{cut}} \right)^3 \right] \quad (8.6)$$

and one for the pressure

$$P^{tail} = \frac{16}{3}\pi\rho^2 \left[\frac{2}{3} \left(\frac{\sigma}{\sigma_{cut}} \right)^9 - \left(\frac{\sigma}{\sigma_{cut}} \right)^3 \right] \quad (8.7)$$

Write a simple Monte Carlo code, following the scheme detailed in class. Perform two sets of simulations, one at reduced temperature $T^* = 2$ (above the critical temperature) and one at $T^* = 0.9$ (well below the critical temperature) and compute, for both cases, the equation of state in the pressure-density plane. Compare the results with the equation of state for the two temperature (data shared).

8.2.1 Solution

The implementation is the usual MC + Metropolis simulation. For each Metropolis sweep we visit each particle in order and propose a move. A proposed move consist on increasing each coordinate by a value $\sim \mathcal{N}[-d_{max}, d_{max}]$ (for all three dim. independently) with $d_{max} = 0.5$ and accepting it with the usual Metropolis rule. To calculate the ΔE of the proposed move, only the pairs between the current particle and all the others are used ($\mathcal{O}(N)$), as opposed to calculate the energy for the whole system ($\mathcal{O}(N^2)$).

The pressure was calculated as:

$$PV = Nk_B T - \frac{1}{3} \sum_i^N \langle \vec{r}_i, \vec{F}_i \rangle = Nk_B T - \frac{1}{3} \sum_{i < j}^N \langle \vec{r}_{ij}, -\frac{dV(r_{ij})}{dr_{ij}} \vec{u}_{ij} \rangle \quad (8.8)$$

$$= Nk_B T - 8\epsilon \left[\left(\frac{\sigma}{r} \right)^6 - 2 \left(\frac{\sigma}{r} \right)^{12} \right] \quad (8.9)$$

The initial position is chosen at random by sampling uniformly inside the box. This could lead to particle starting very close with $V_{LJ}(r)$ very high and pressures very high. To avoid including these ludicrous values, we simply discarded the first 500 values of the pressure when calculating the mean. To improve performance, the distance is always calculated squared. There is no need to calculate the square root of it because since it is monotone, every comparison can be done on the squared values. Also in every formula the distance is always to an even power, so if we have r^2 , then $\left(\frac{\sigma}{r} \right)^6 = \left(\frac{\sigma^2}{r^2} \right)^3$.

For every point of pressure ρ given in the data shared, we set $N = 200$ and we calculate the corresponding volume with $V = N/\rho$. From which $\sigma_{cut} = \sqrt[3]{V}/2$. The final results are shown in figure 8.1.

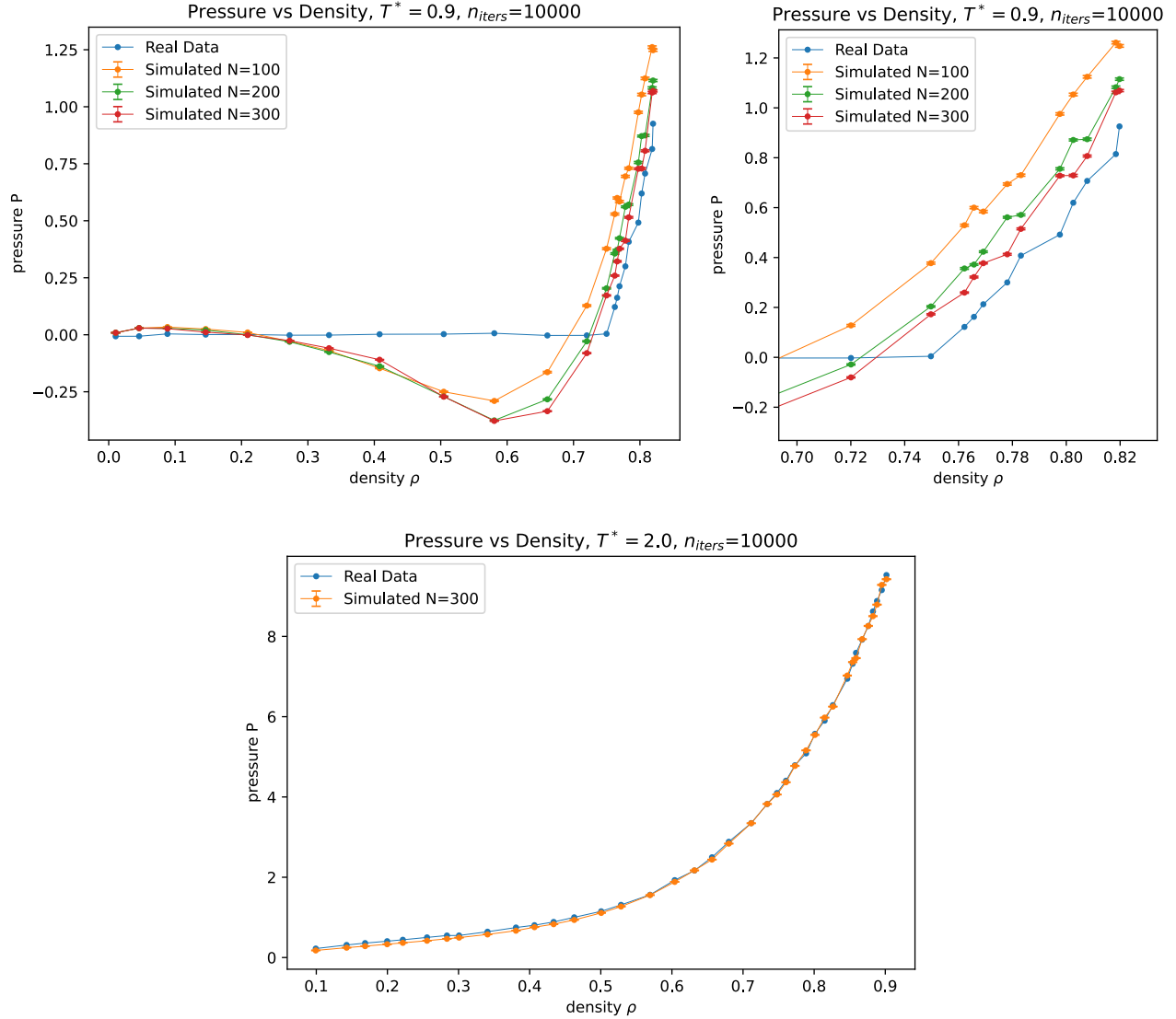


Figure 8.1: The pressure versus density plots. The first row shows the case for $T^* = 0.9$ with a zoomed in version, the second row the case for $T^* = 2$. The former case has different lines for different N because we wanted to show if the situation could be improved with more particles, instead for the latter case there is no need because the data is already in excellent agreement.

In the first row of the figure, with $T^* = 0.9$, the data simulated do not agree at all with the data shared, there is even a region with negative pressure. Increasing N does not reduce this region but makes the data closer to the real data in the last area. In the second figure instead, with $T^* = 2$, there is excellent agreement (only qualitatively because the low errors $\sqrt{\sigma^2/N}$ are relatively far from the real points). This makes us think that the problem with the first figure it's not caused by an error in the simulation code, but probably that our simple simulation cannot capture the complex phenomena happening with $T < T_c$. Presumably, when $T > T_c$ these phenomena do not happen at all instead of just being less energetic than $k_B T$, so these phenomena are not present at all when we are over the T_c . We think that because a phase transition is usually a transition from something "structured" to something "unstructured".

Exercises Lecture 9

9.1 Integration schemes

The harmonic oscillator Implement the Velocity Verlet algorithm and the predictor-corrector algorithm. Consider the harmonic oscillator

$$\frac{dq}{dt} = p, \quad \frac{dp}{dt} = -\omega^2 q. \quad (9.1)$$

The exact solution of $p(t)$ vs $\omega q(t)$ we get a circular orbit in phase that rotates by an amount of $\omega\Delta t$ at every time step. Considering this system, compare the two algorithms implemented above in terms of i) energy conservation, plotting $(E(t) - E_0)/E_0$ vs t , E_0 being the initial total energy ii) discrepancy with the analytical solution as a function of time. Verify that the Velocity Verlet is stable (i.e. it follows the analytical solution) for $\omega\Delta t < 2$ checking a few values of ω (e.g. $\omega = 0.01, 1, 100$).

9.1.1 Solution

Since:

$$\frac{d^2 q}{dt^2} = \frac{dp}{dt} = -\omega^2 q \implies q(t) = A \sin(\omega t + \varphi) \quad (9.2)$$

$$p = \frac{dq}{dt} \implies p(t) = A\omega \cos(\omega t + \varphi) \quad (9.3)$$

with A and φ chosen by setting the initial conditions:

$$q(t_0 = 0) \stackrel{!}{=} 0 \implies q(0) = A \sin(\varphi) = 0 \implies \varphi = 0 \quad (9.4)$$

$$p(t_0 = 0) \stackrel{!}{=} 1 \implies p(0) = A\omega \cos(0) = 1 \implies A = \frac{1}{\omega} \quad (9.5)$$

We implemented the Velocity Verlet algorithm and the predictor-corrector Beeman's algorithm. We first plotted, for both methods, the difference in energy with the exact energy as a function of time to see if the energy is conserved. The results are shown in figure [9.1](#).

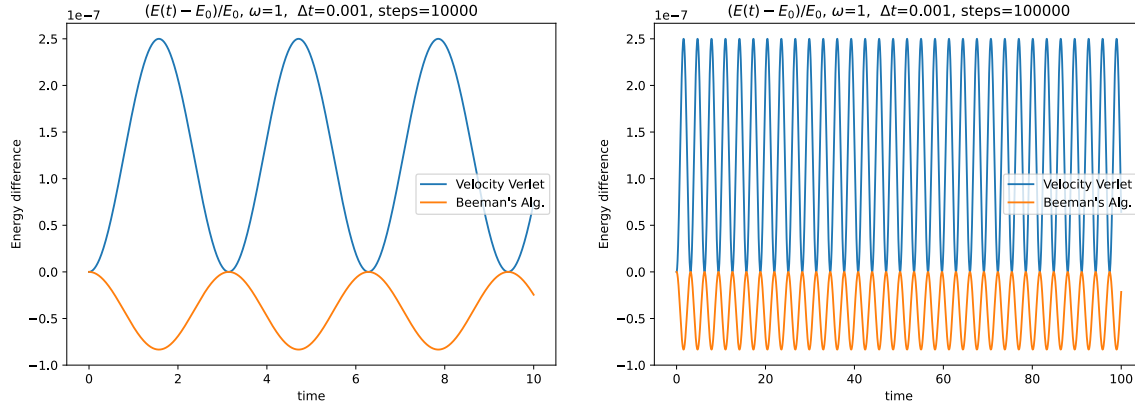


Figure 9.1: The discrepancy between the two methods as function of time.

The energy is conserved after 100000 iterations and the magnitude of the error is the same for both algorithm. If we instead compare the $q(t)$ and $p(t)$ with the analytical solution (by plotting the difference between the two) we obtain the graphs in figure 9.2.

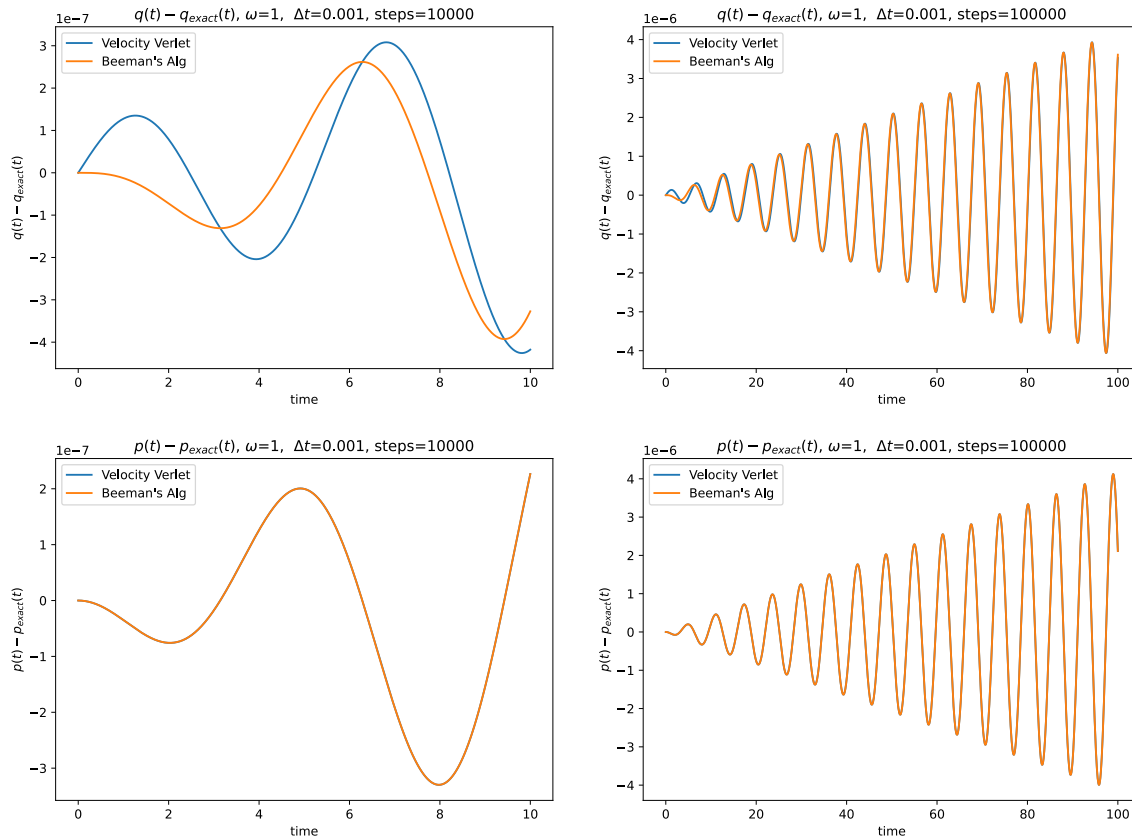


Figure 9.2: Comparing $q(t)$ and $p(t)$ with the analytical value, with two different timescales.

For $q(t)$ the error is a bit different for the two methods, but then they overlap pretty nicely in all the others graphs. We can see that the errors is slowly building up as we increase the number of steps.

Lastly we check how stable is the solution of the Velocity Verlet as $\omega\Delta t$ get close to 2. To to that, we plot the graph $p(t)$ versus $q(t)$. Analytically, with the exact solution, we have (remember that

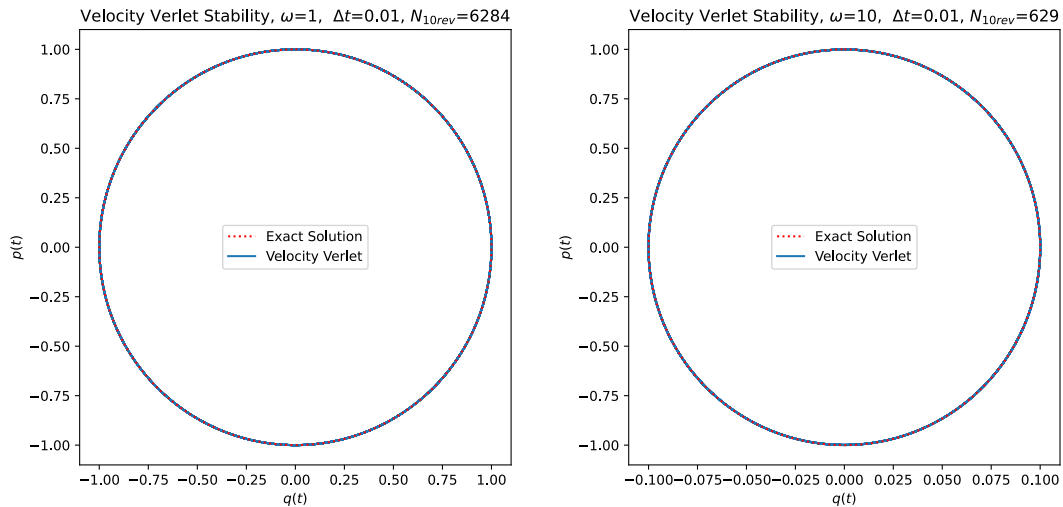
$A = 1/\omega$:

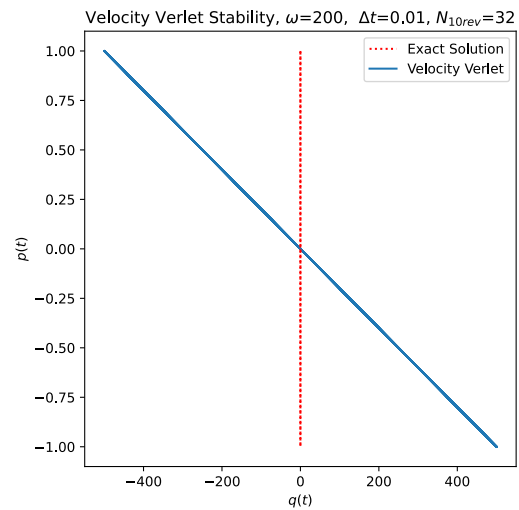
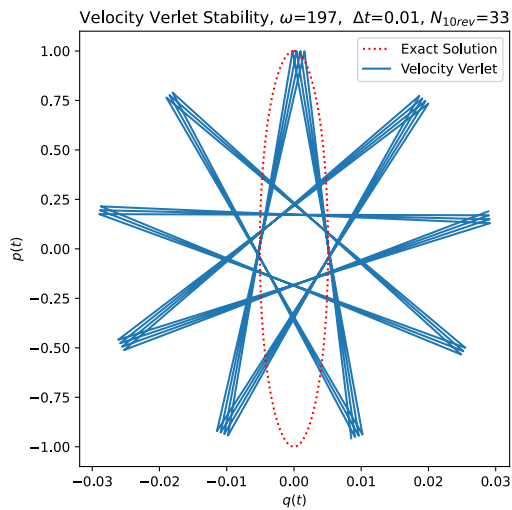
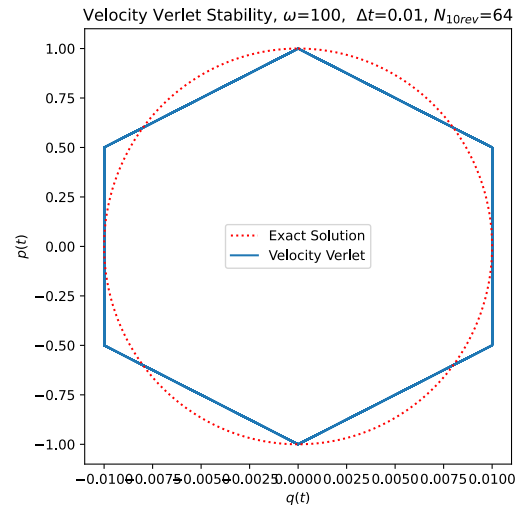
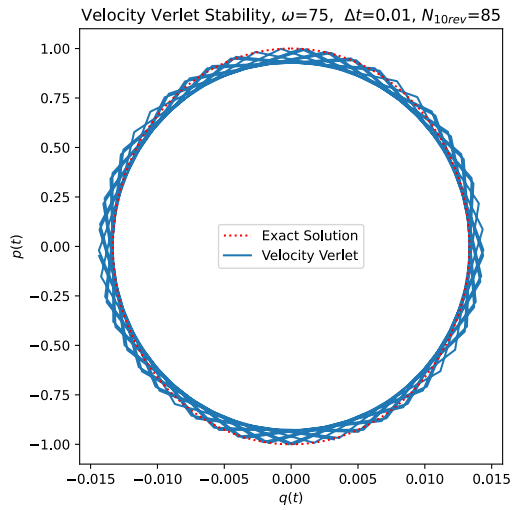
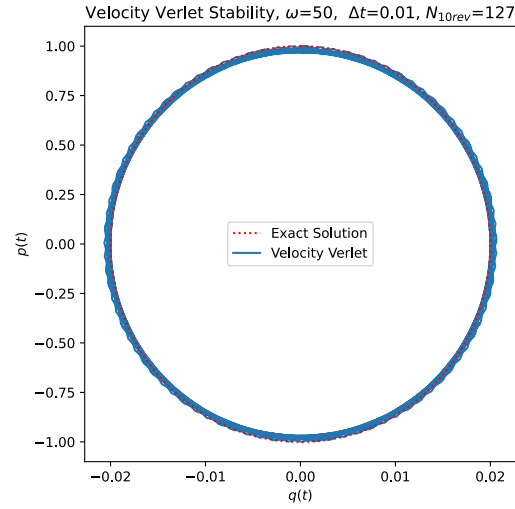
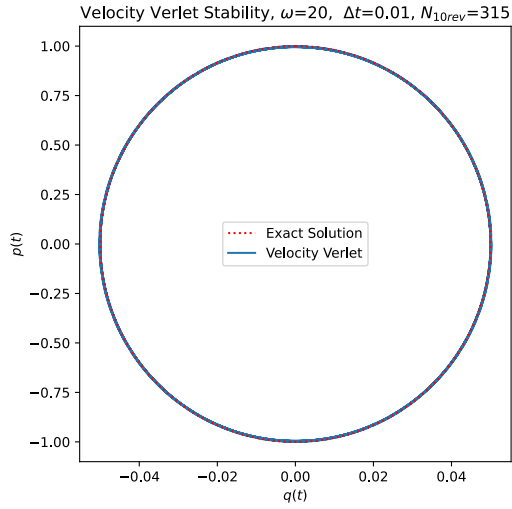
$$q(t) = \frac{1}{\omega} \sin(\omega t) \quad p(t) = \cos(\omega t) \implies q(t)^2 + p(t)^2 \text{ is an ellipse with radius } r_x = 1/\omega, r_y = 1 \quad (9.6)$$

so we compare the shape of the algorithm with the ellipse. To avoid plotting all the path we plot the last 10 revolutions of the Velocity Verlet (and the ellipse). A revolution is when ωt ranges from 0 to 2π , so if we want the last 10 revolutions, we just plot the last N_{10rev} points:

$$N_{10rev} = 10 \cdot \frac{2\pi}{\omega} \cdot \frac{1}{\Delta t} \quad (9.7)$$

the results are shown in the following figures (the algorithm ran for 50000 iterations and we are looking at the last N_{10rev}). From the figures, we can see that until $\omega = 20$ and so $\omega\Delta t = 0.2$, the shapes are the same as the analytical solution. From $\omega = 50$ ($\omega\Delta t = 0.5$) we can see the trajectories going a bit off, the Velocity Verlet algorithm is calculating the positions and the velocities for timestep too big so it cannot follow smoothly the circle and some little spikes appear. At $\omega = 100$ ($\omega\Delta t = 1$) the timestep is so big that in 6 iteration goes around the circle: the angle between the lines is getting closer and closer to 0, but the points are still bounded. At $\omega = 200$ ($\omega\Delta t = 2$) the algorithm just bounces between two points. With any $\omega > 200$ the algorithm quickly overflows to infinity, and it's hard to plot anything meaningful.





Exercises Lecture 10

10.1 Interaction potentials & thermostats - Canonical fluctuations (pen&paper)

Show that the fluctuations of the temperature of the system, as defined in class, in the canonical ensemble are

$$\frac{\sigma_{T_K}^2}{\langle T_K \rangle^2} = \frac{\langle T_K^2 \rangle - \langle T_K \rangle^2}{\langle T_K \rangle^2} = \frac{2}{3N}$$

10.1.1 Solution

Since we know that:

$$T_k = \frac{1}{3Nk_B} \sum_i m_i v_i^2 = \frac{2E_k}{3Nk_B} \implies \langle T_k \rangle = \frac{2\langle E_k \rangle}{3Nk_B} \quad \text{and} \quad \sigma_O^2 = \langle O^2 \rangle - \langle O \rangle^2 \quad (10.1)$$

then

$$\frac{\sigma_{T_K}^2}{\langle T_K \rangle^2} = \frac{4}{9N^2k_B^2} \frac{\langle E_k^2 \rangle - \langle E_k \rangle^2}{\langle T_K \rangle^2} \quad (10.2)$$

now, since we are in the canonical ensemble, we know that

$$\sigma_E^2 = \langle E_k^2 \rangle - \langle E_k \rangle^2 = k_B T^2 \frac{\partial \langle E_k \rangle}{\partial T} \quad (10.3)$$

so then

$$\frac{\sigma_{T_K}^2}{\langle T_K \rangle^2} = \frac{4}{9N^2k_B^2} \frac{k_B T_K^2 \frac{\partial \langle E_k \rangle}{\partial T_K}}{\langle T_K \rangle^2} = \frac{4}{9N^2k_B^2} \frac{3Nk_B^2 T_K^2}{2T_K^2} = \frac{2}{3N} \quad (10.4)$$

10.2 Interaction potentials & thermostats - Lennard-Jones fluid in different ensembles

Note: This exercise can be carried out with LAMMPS.

Simulate N particles of mass $m = 1$ in a cubic box of length L (with periodic boundary conditions). The particles interact with each other through a Lennard-Jones (LJ) potential

$$V_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

where $\sigma = 1$ and $\epsilon = 1$. Choose $L = 10\sigma$ and $\rho = N/V = 0.2\sigma^{-3}$. Integrate the equations of motion with the Velocity Verlet algorithm.

Generate a suitable initial condition. Choose one of the two strategies introduced in one of the previous exercises. Draw the initial velocity distribution from the equilibrium (Maxwell-Boltzmann) distribution at $T^* = 1$, enforcing the total momentum to zero.

- **NVE ensemble:** Simulate the system without a thermostat. Check the effect of the cut-off radius: perform different simulations, increasing the cut-off from $r_c = 2^{1/6}\sigma$ to $r_c = 4\sigma$ with steps $\Delta r_c = 0.2\sigma$ (round the numbers to the first decimal place after the first one). After equilibration (if needed), compute the radial distribution function to compare the different cases.
- **NVT ensemble:** Introduce now a thermostat, picking one non-canonical (Velocity rescaling or Berendsen) and one canonical (Andersen or Nose-Hoover - Note: the latter can be done in LAMMPS). Simulate again the same LJ system as in the previous exercise. Fix the reference temperature of the heat bath $T^* = 2$. Integrate the equations of motion with the Velocity Verlet algorithm. Compute the kinetic energy and verify that, in both cases, the average is consistent with the equipartition theorem ($\langle E_K \rangle = 3/2 N k_B T$ upon varying the number of particles in the system (keeping the box size fixed); keep $\rho \leq 0.2$. Check that the fluctuations of the temperature are consistent/not consistent with the canonical ensemble upon varying the number of particles in the system as above.

10.2.1 Solution

Since the density and the volume are specified, this set $N = 200$. For the initial configuration of the positions, we put 20 particles for each layer of z (with $\Delta z = 0, 1\sigma, 2\sigma, \dots$). In each layer, the 20 particles are in a rectangle 4×5 with $\Delta x = \Delta y = 2\sigma$ and starting in $(2\sigma, 1\sigma)$. We did that because we did not want random positions that could lead to very close distances and so very high velocities that could lead to instability, so we spread them out like a lattice. The velocity vector components are sampled from $\mathcal{N}(0, \sqrt{T})$, so that the norm of those vectors follows the Maxwell-Boltzmann distribution (with $T = 1$). All the simulation were done with $\Delta t = 10^{-2}$ and with 10^5 iterations.

Microcanonical ensemble (NVE) To calculate the pair correlation function $g(r)$ we used:

$$g(r) = \frac{1}{4\pi r^2} \frac{V}{N^2} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} \delta(r - r_{ij}) \approx \frac{1}{4\pi r^2} \frac{V}{N^2} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} \frac{1}{\sqrt{2\pi}\sigma_g} \exp\left(-\frac{1}{2} \frac{(r - r_{ij})^2}{\sigma_g^2}\right) \quad (10.5)$$

For approximating δ we used a gaussian centered in r with $\sigma_g = 0.02$. The values for $g(r)$ are then averaged over the last 50 iterations to smooth them out. We plot the results of the simulation of this ensemble in figure 10.1 and 10.2 with different values of the cutoff value σ_c .

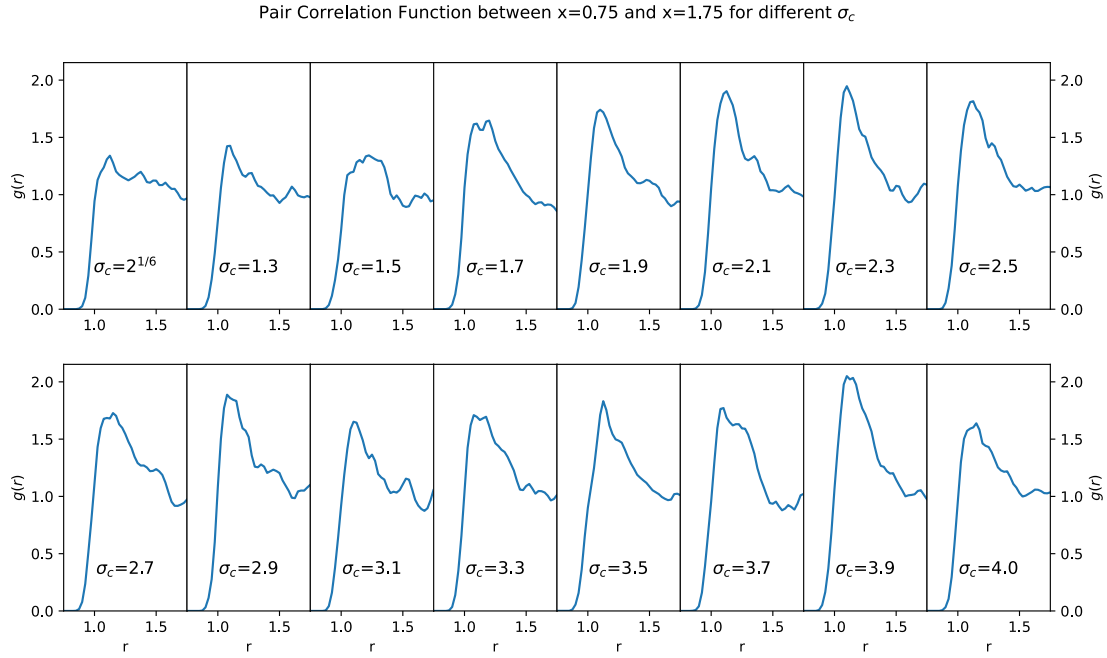


Figure 10.1: Pair correlation function $g(r)$ for different values of σ_c in the NVE ensemble. Here are shown only the points between $r = 0.75$ and $r = 1.75$.

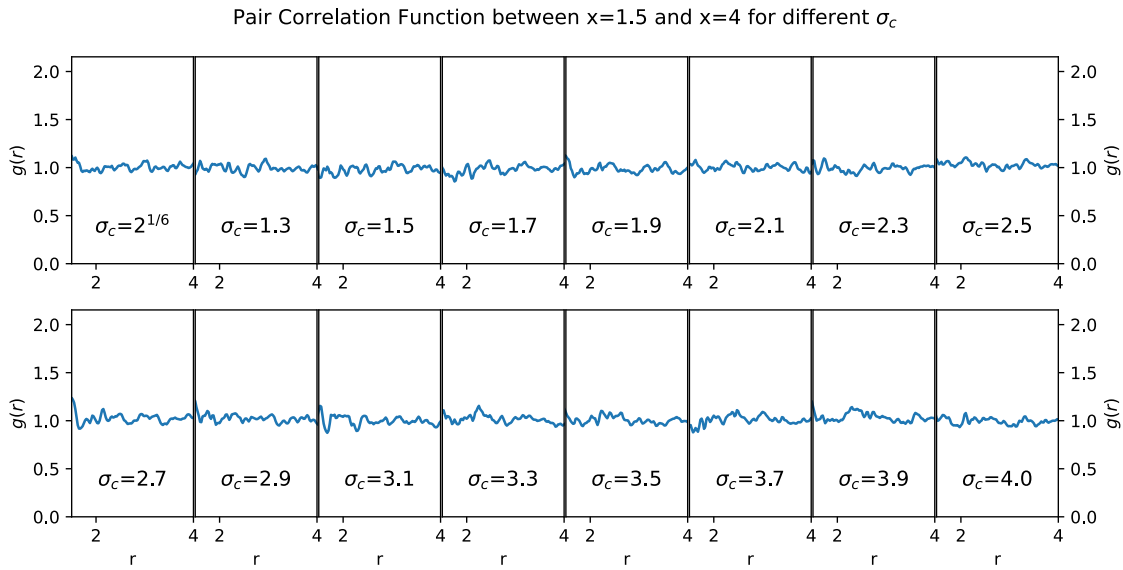


Figure 10.2: Pair correlation function $g(r)$ for different values of σ_c in the NVE ensemble. Here are shown only the points between $r = 1.5$ and $r = 4$.

First we note the presence of a peak in $r = 2^{1/6}$, the minimum of the potential, which means that particles tend to cluster with an average distance of $2^{1/6}$. This peak is not present when σ_c is set to $2^{1/6}\sigma$. This is expected because we are not considering anything after the minimum of the potential, so there is only the repulsive part and no incentive for the particles to bond together. As soon as we increase the cutoff, the peak starts to appear because now the potential has a proper minimum with

an attractive and repulsive part. Also there seems to be an oscillating behavior with the peaks getting higher at around $\sigma_c = 2.2$, 2.7 and 3.5 . After the peak, the function is the same for all the cutoffs fluctuating around 1.

Canonical ensemble (NVT) For this simulation we used the velocity rescaling and the Andersen thermostat. We still initialized the particles as before: even if N is changing we put the particles in the same lattice-like positions as before, starting filling them from the z-layer 0 and up until we ran out of particles. As cutoff we choose $\sigma_c = 4$. We kept the volume the same and we just decrease the number of particles (so $\rho \leq 0.2$).

For the velocity rescaling, the results are in table 10.1. As expected there is perfect agreement between the theoretical and the simulated kinetic energy, since we are rescaling the velocities to have the desired temperatures. Also the fluctuations of the temperature are 0 because as before we are rescaling the velocities to have that particular temperature.

NVT Ensemble with Velocity Rescaling at $T^* = 2$				
N	$\langle E_k^{th} \rangle$	$\left(\frac{\sigma_{T_k}^2}{\langle T_k \rangle^2} \right)^{th}$	$\langle E_k^{sim} \rangle$	$\left(\frac{\sigma_{T_k}^2}{\langle T_k \rangle^2} \right)^{sim}$
25	75	0.027	75.0	$< 10^{-10}$
50	150	0.013	150.0	$< 10^{-10}$
75	225	0.0089	225.0	$< 10^{-10}$
100	300	0.0067	300.0	$< 10^{-10}$
125	375	0.0053	375.0	$< 10^{-10}$
150	450	0.0044	450.0	$< 10^{-10}$
175	525	0.0038	525.0	$< 10^{-10}$
200	600	0.0033	600.0	$< 10^{-10}$

Table 10.1: Results for the NVT Ensemble with the Velocity rescaling thermostat

Instead for the Andersen thermostat, we used $\omega = 0.05/\Delta t$ because then the probability to reset the velocity of a particle is $p = \omega\Delta t = 0.05$. So on average 5% of the particles are chosen to have its velocity sampled from the Maxwell-Boltzmann distribution with $T^* = 2$. The results are shown in table 10.2. The results are in good agreement with the theoretical ones. Also in this case there are fluctuations since we are setting the velocities of only 5% of the particles, the others are left free to interact and change its energy.

NVT Ensemble with Andersen Thermostat at $T^* = 2$				
N	$\langle E_k^{th} \rangle$	$\left(\frac{\sigma_{T_k}^2}{\langle T_k \rangle^2} \right)^{th}$	$\langle E_k^{sim} \rangle$	$\left(\frac{\sigma_{T_k}^2}{\langle T_k \rangle^2} \right)^{sim}$
25	75	0.027	75.11	0.026
50	150	0.013	150.0	0.013
75	225	0.0089	225.1	0.0091
100	300	0.0067	300.0	0.0067
125	375	0.0053	375.3	0.0054
150	450	0.0044	450.4	0.0044
175	525	0.0038	525.9	0.0038
200	600	0.0033	599.1	0.0034

Table 10.2: Results for the NVT Ensemble with the Andersen thermostat

Exercises Lecture 11

11.1 Langevin and Brownian Dynamics - Brownian time and length scales (pen&paper)

- In the overdamped limit, the diffusion scale becomes the relevant time scale. Using the Stokes-Einstein formula, estimate the time scale of a particle of diameter $\sigma = 10^{-8}$ m to diffuse in water over a distance equal to its own diameter. Is this time scale compatible with the requirement of the Brownian description? (Note: the correlation time of water molecules is of the order of picoseconds, take that as the time scale of the fluctuations τ_{coll})
- Conversely, show that, above $\sigma = 5 \mu\text{m}$, Brownian motion becomes negligible. Which other force becomes relevant?

(*Hint:* The energy scale is always the thermal energy at room temperature; for simplicity, the object has roughly the same density as water; also, the object is spherical.)

Solution

The Stokes-Einstein formula is:

$$D = \frac{k_B T}{6\pi\eta r} = \frac{1.381 \cdot 10^{-23} \text{ J/K} \cdot 293 \text{ K}}{6\pi (10^{-3} \text{ Pa} \cdot \text{s})(0.5 \cdot 10^{-8} \text{ m})} = 4.293 \cdot 10^{-11} \frac{\text{m}^2}{\text{s}} \quad (11.1)$$

where η is the viscosity of water at room temperature, $r = \sigma/2$ and $T = 293$ K. In the Langevin overdamped case, we have $\langle x^2 \rangle = 2Dt$. If we want to know the timescale for the particle to diffuse to a distance of σ :

$$t = \frac{\langle \sigma^2 \rangle}{2D} \implies t_{diff} = \frac{\sigma^2}{2D} = 1.165 \cdot 10^{-6} \text{ s} \quad (11.2)$$

Since $t_{diff} \gg \tau_{coll} \sim 10^{-12}$ s then the time scale is compatible with the requirement of the Brownian description. If the particle starts to be bigger, its mass starts to be relevant. If the particle is diffusing in 3D and the gravitational pull it's much stronger than the thermal energy then the particle will just sink below (assuming $\rho_{particle} > \rho_{water}$). Let's calculate the change in energy if the particle moves down by σ :

$$\Delta E = mgh = V\rho g\sigma = \frac{4\pi(\sigma/2)^3}{3} \cdot 10^3 \frac{\text{kg}}{\text{m}^3} \cdot 9.81 \frac{\text{m}}{\text{s}^2} \sigma = 3.21 \cdot 10^{-18} \text{ J} \quad (11.3)$$

which compared to the thermal energy $k_B T \approx 4 \cdot 10^{-21}$ J is three order of magnitude bigger.

11.2 Langevin and Brownian Dynamics - Brownian motion

Consider an isolated particle (or an “ideal gas” of non interacting Brownian particles). Set $m = 1$, $\sigma = 1$ and $\epsilon = 1$ as units of mass, length, energy. Set $L = 20\sigma$ and apply periodic boundary conditions. Choose initial conditions as you prefer; the most convenient choice is to set all the starting positions in the origin.

Brownian motion in the bulk:

Simulate this system in the overdamped limit (first order integrator) or in the “underdamped” limit (Stochastic Velocity Verlet or with the second order integrator). Compute the mean square displacement and discuss what happens upon varying the temperature ($0.1 < T^* < 2$, $\gamma = 1\tau^{-1}$) and the friction coefficient ($0.1 < \gamma\tau < 100$, $T^* = 1$). Pick 5-10 values in each case. Compare the results with the theory. Further measure the distribution of the displacement of the particle(s) along one axis at different times: what changes to the distribution as time progresses?

Brownian motion in an harmonic trap:

Consider single overdamped particle (or, again, an ideal gas of non-interacting overdamped particles) in an harmonic trap, centred in a certain point of the box $\vec{r}_c$ (for simplicity, choose the origin of the reference frame or the centre of the box).

$$V(r) = \frac{1}{2}K(\vec{r} - \vec{r}_0)^2$$

Set $L = 20\sigma$ and apply periodic boundary conditions. Fix the initial conditions in the centre of the trap. Compute the average and the variance of the position along one axis as a function of $0.1 < K\sigma^2/\epsilon < 10$, $0.1 < \gamma\tau < 100$, $0.1 < T^* < 2$ (Note: when varying one quantity, keep the others fixed to unitary value). Measure both the mean square displacement and the displacement distribution along one axis. Discuss the result, in comparison to the bulk case.

Solution

We simulated the system in 2D with $L = 20\sigma$, periodic boundary conditions, origin in $O = (0, 0)$ and sides extending from 0 to L . The simulations consisted of $N = 5000$ non-interacting particles in the overdamped (first order integrator) case, with a $\Delta t = 0.01$ and 3000 steps.

Brownian motion in bulk

In this case we simulated the system in the overdamped limit with 5 different T^* and 5 different $\gamma\tau$. To calculate the mean square displacement while using the periodic boundary condition (PBC), we decide to use the (signed) distance, for each axis, with respect to the origin. Using the raw value of the position introduce discontinuity; going a bit to the left or down from the origin make the position changes $\approx L$. Instead by using the distance (squared) we solve this problem, the extreme values are then $\pm L/2$ (the point $L - \varepsilon$ becomes $-\varepsilon$). From now on “mean square displacement” means this.

In figure 11.1 we show the mean squared displacement (from the origin) at different times for the different temperatures (with $\gamma\tau = 1$). Each point is the average values of all the particles at that time with that value of temperature. The results are specifically for the x-axis, even if the system was simulated in 2D, but the results are the same for each axis (stochastically speaking), as it should since they are completely independent. We also show, as dotted lines, the theoretical values in the overdamped limit: $\langle (x(t) - x_0)^2 \rangle = \frac{2k_B T}{m\gamma} t$. At the start, the curves follow exactly the theoretical values, they grow linearly, and this is expected because our integrator is for the overdamped limit. The particles with higher temperatures tend to move longer in a timestep and thus reaching positions more further away. Later they diverge from the theoretical solution and reach a plateau, this is also expected since our system has periodic boundary conditions. After $x = L/2$ they start to return to the original position and the system will reach equilibrium around a characteristic value (we suppose) related to the size of the system.

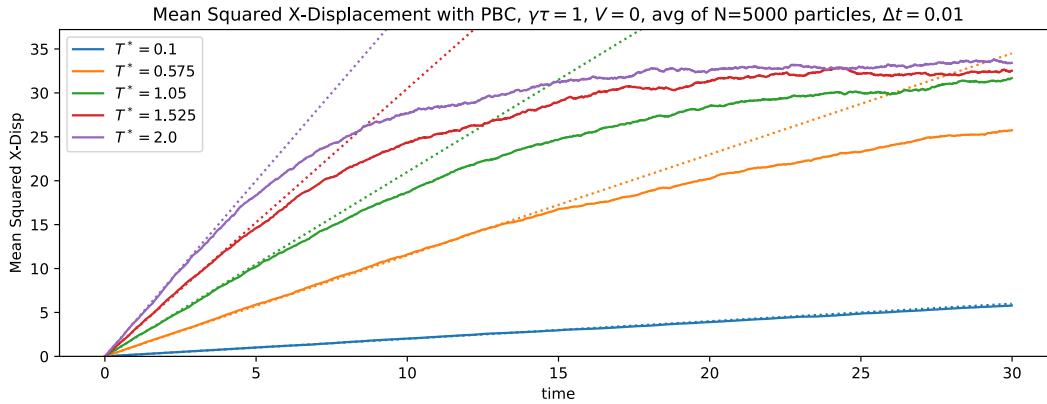


Figure 11.1: Mean squared displacement (calculated using the distance from the origin) for different temperatures with $\gamma\tau$ fixed. There is no potential here. The dotted line is the theoretical one (for the overdamped limit, without PBC).

We also show the distribution of the displacement (not squared) at four different time for all the temperatures in figure 11.2. Precisely, the histogram contains the data at that particular time of all the N particles at that temperature. We can recognize the shape of a gaussian distribution where lower temperatures mean less variance and higher peaks, but as time increase, because of the PBC, the positions tend to be more and more uniformly distributed and occupy the space evenly.

Now we show the same graphs but keeping the temperature constant ($T^* = 1$) but changing $\gamma\tau$. The results are in figure 11.3 and 11.4. The results are the same as with the temperature except that lower $\gamma\tau$ means bigger steps. In this case without any potential ($F = 0$), the results of $\gamma\tau$ are the same as with $T^* = \frac{1}{\gamma\tau}$ because of the first order integrator formula:

$$r(t + \Delta t) = r(t) + \frac{F(r(t))}{m\gamma} \Delta t + \sqrt{\frac{2k_B T}{m\gamma}} \xi \sqrt{\Delta t} \quad (11.4)$$

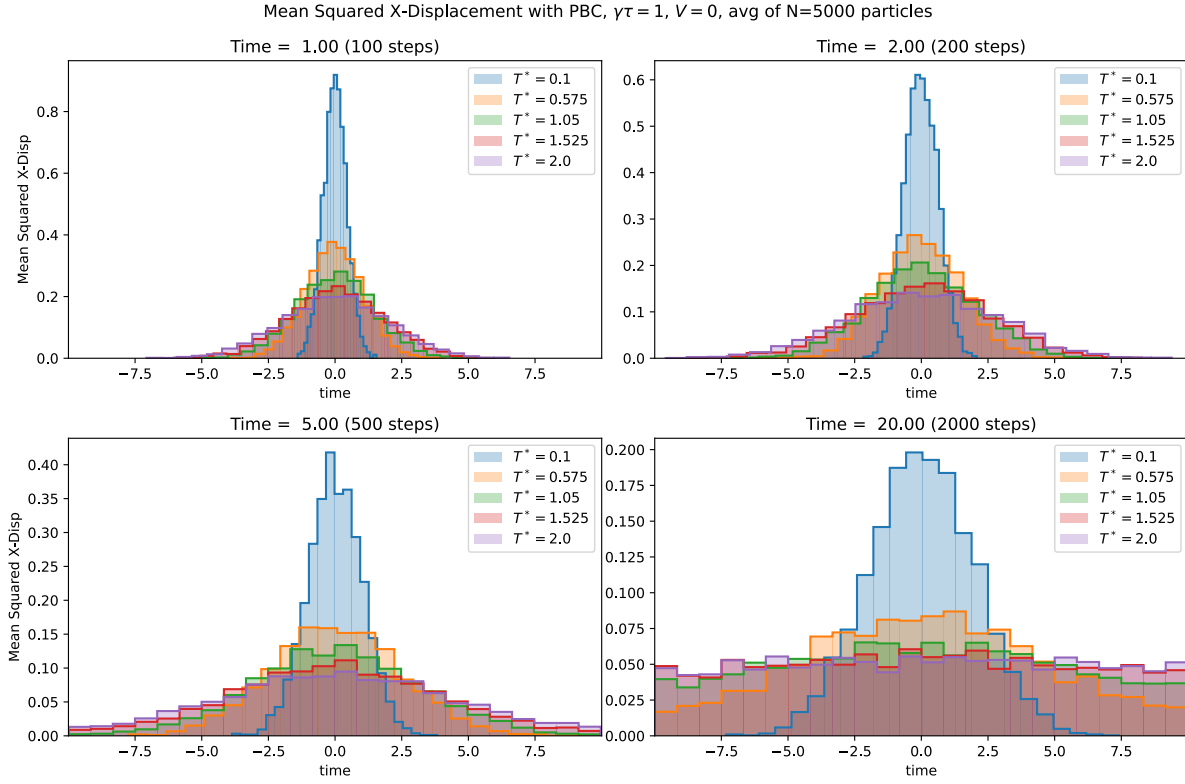


Figure 11.2: Distribution of the mean squared displacement for different temperatures with $\gamma\tau$ fixed for different times. There is no potential here. As time passes, because of the PBC, the distribution tend to flatten and visit every point uniformly.

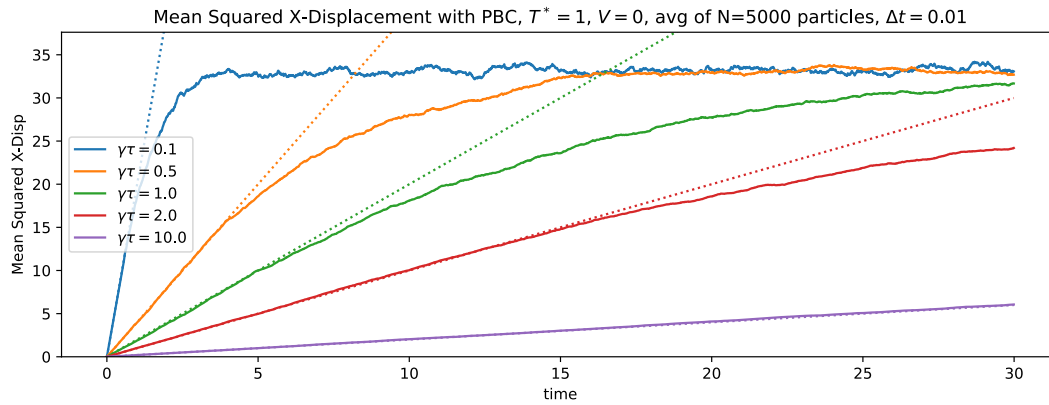


Figure 11.3: Mean squared displacement for different $\gamma\tau$'s with temperature fixed. There is no potential here. The dotted line is the theoretical one (for the overdamped limit, without PBC).

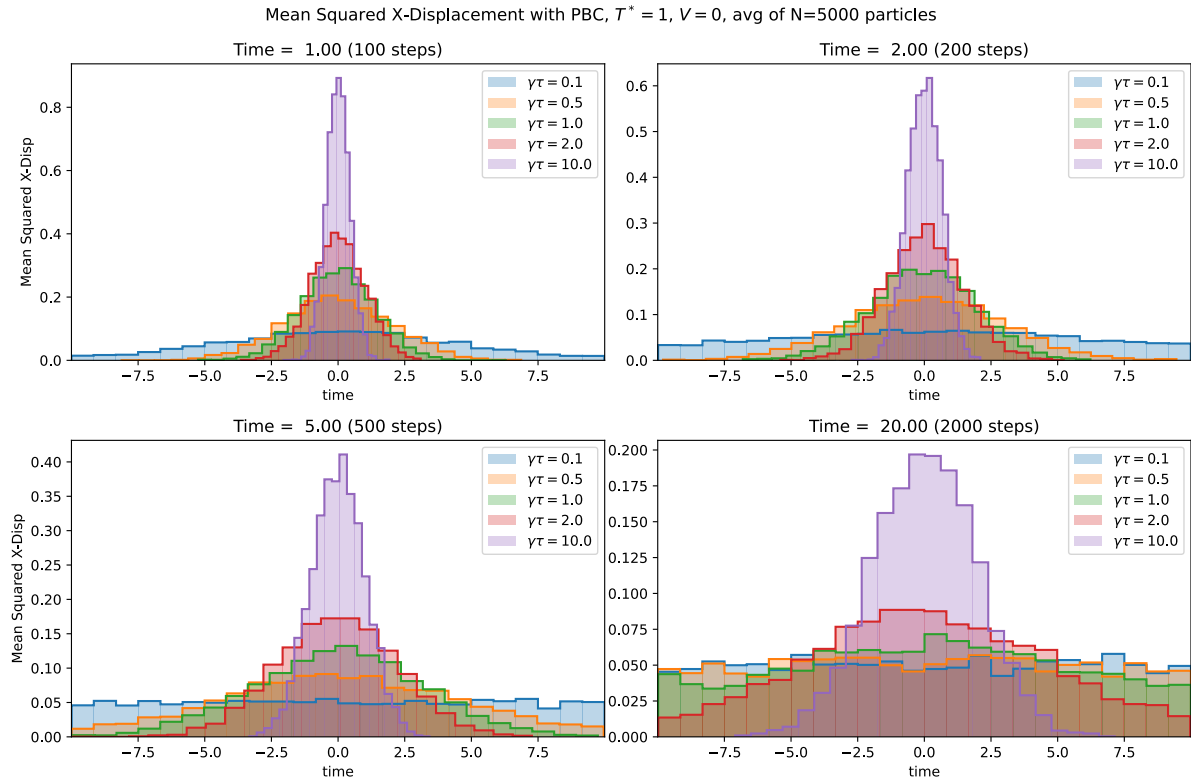


Figure 11.4: Distribution of the mean squared displacement for different $\gamma\tau$'s with temperature fixed for different times. There is no potential here. As time passes, because of the PBC, the distribution tend to flatten and visit every point uniformly.

Brownian motion in an harmonic trap

In this case the setup is the same as before except now there is an harmonic potential with center in $(0,0)$ which is also the starting position of the particles. So, for different values of the temperatures, $\gamma\tau$'s and K 's we calculate the average and the variance of the position along one axis. We report the results in table 11.1.

System with harmonic trap, x-position								
$K = 1$ and $\gamma\tau = 1$			$K = 1$ and $T^* = 1$			$T^* = 1$ and $\gamma\tau = 1$		
T^*	$\langle x \rangle$	σ_x	$\gamma\tau$	$\langle x \rangle$	σ_x	K	$\langle x \rangle$	σ_x
0.1	-0.000782	0.0982	0.1	0.00123	1.05	0.1	0.00175	8.21
0.575	-0.00314	0.572	0.5	0.00239	1.00	0.2	-0.0190	4.53
1.05	-0.00747	1.03	1	0.00527	0.989	0.5	0.0111	1.95
1.525	0.000610	1.51	2	-0.00376	0.972	1	0.00342	0.993
2.0	-0.006862	1.98	10	0.0140	0.841	10	-0.000234	0.105

Table 11.1: The table shows the average and the variance of the

The mean value it's always around zero as expected since that's where the center of potential is located and by plotting the position (not squared) we expected random displacements around the potential with mean zero. Instead, the variance changes a lot based on the value of the parameters, high temperatures, low γ s and low K s mean higher variance. This is also expected because respectively higher energy, lower friction and lower recalling force make the particles roam more, the opposite instead make the particles locked in place. This behaviors can also be seen in the images of the mean square displacement and its distribution over time (along one axis) for all three parameters in the following figures.

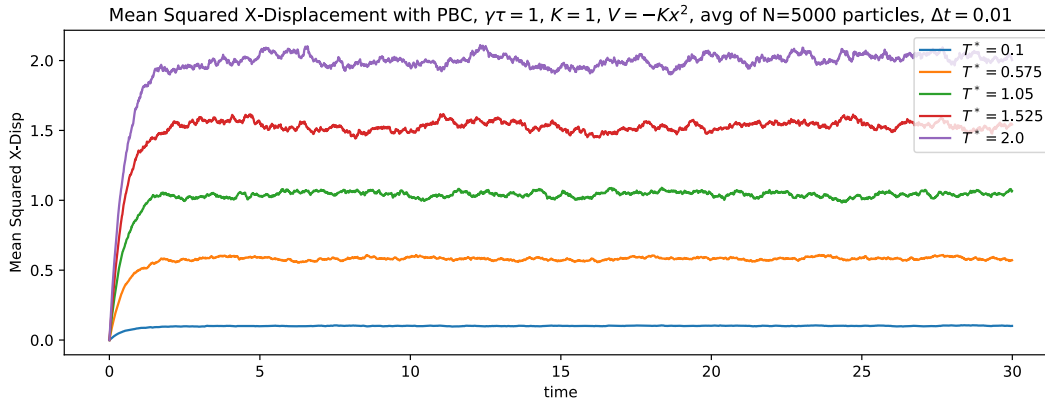


Figure 11.5: Mean squared displacement for different temperatures with $\gamma\tau$ and K fixed.

In case where the temperature changes (figure 11.5 and 11.6), the recalling harmonic force even with $K = 1$ is very strong. In fact, comparing these images with the one of the previous case, we see that the particles moves a lot less, reaching plateau values a lot smaller than before. This also mean that the distribution does not feel the PBC at all and it stays the same through time.

Instead if we change the friction coefficient (figure 11.7 and 11.8), the harmonic trap keeps the particles at a fixed distance, but higher friction values make the reaching of that stable distance harder. The particles with higher friction coefficient are harder to move, but given enough time they

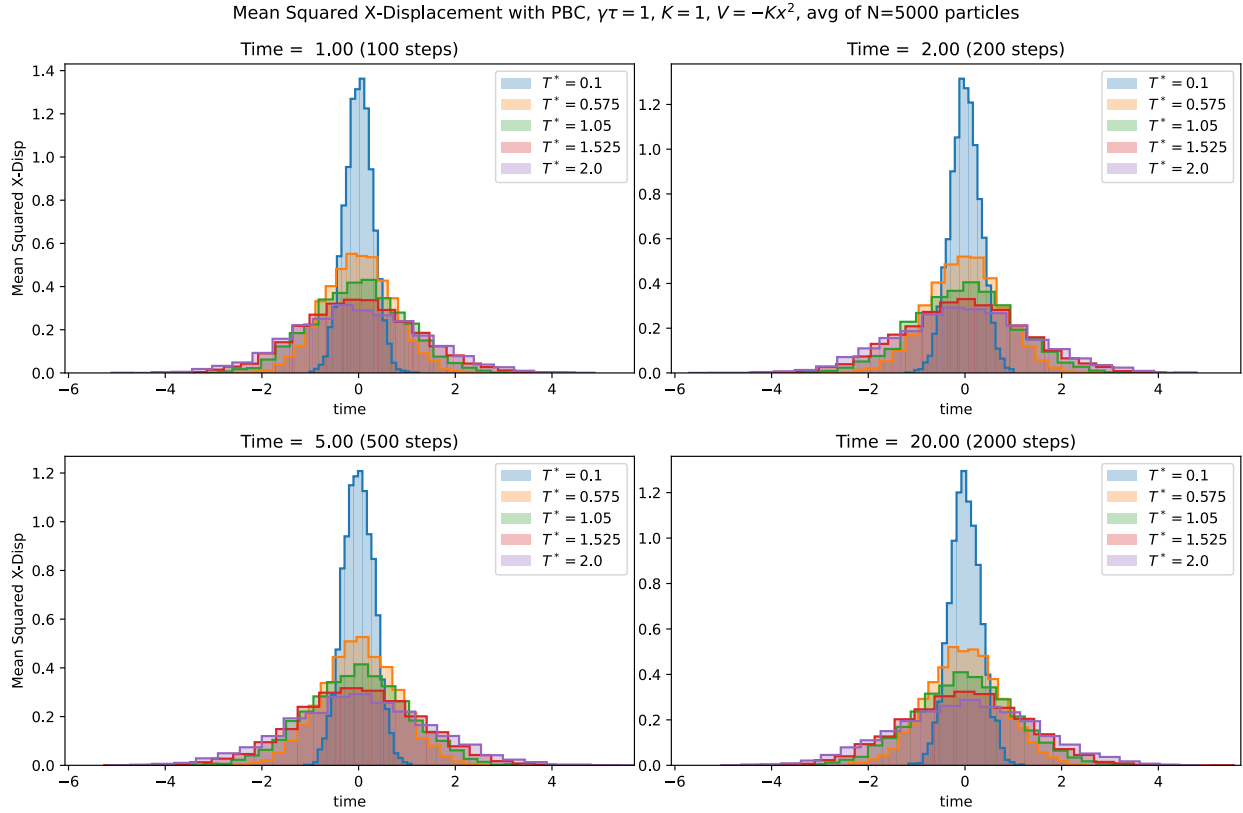


Figure 11.6: Distribution of the mean squared displacement for different $\gamma\tau$'s and K 's with temperature fixed for different times. As time passes, even of the PBC, the distribution stays the same because the particles are confined in a small region around the potential.

also stabilize around the same distance from the center of the potential and match the distribution of the others.

Lastly, by changing the recalling force coefficient (figure 11.9 and 11.10) make the the equilibrium point around the center of the potential different. Higher pull, smaller distance of attraction. Also in this case the PBC does not change the outcomes. In fact we can see that the average distance squared reached by the case with $K = 0.1$ is still far less than the ones reached when $V = 0$.

An animation can be found on [GitHub](#).

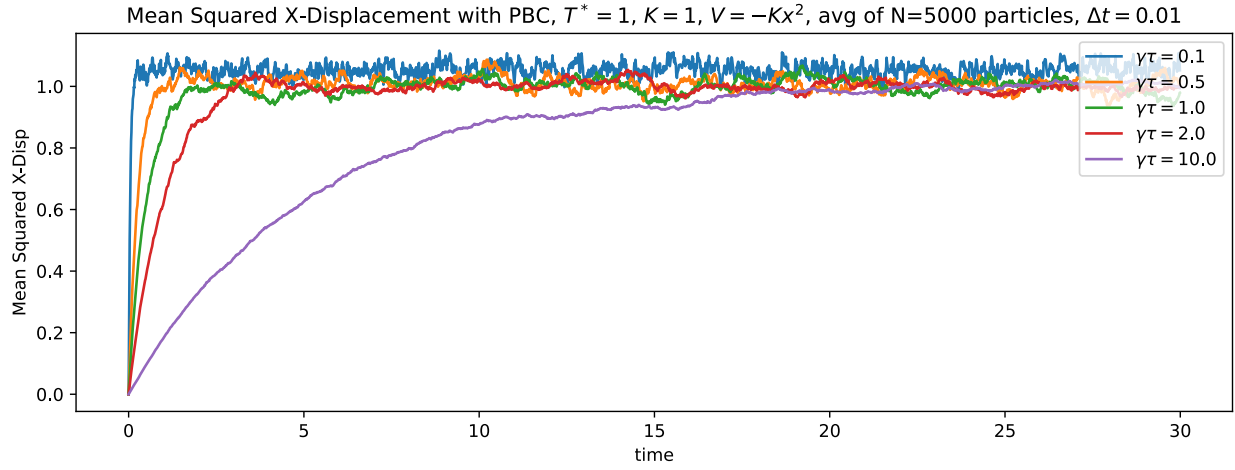


Figure 11.7: Mean squared displacement for different $\gamma\tau$ s with T and K fixed.

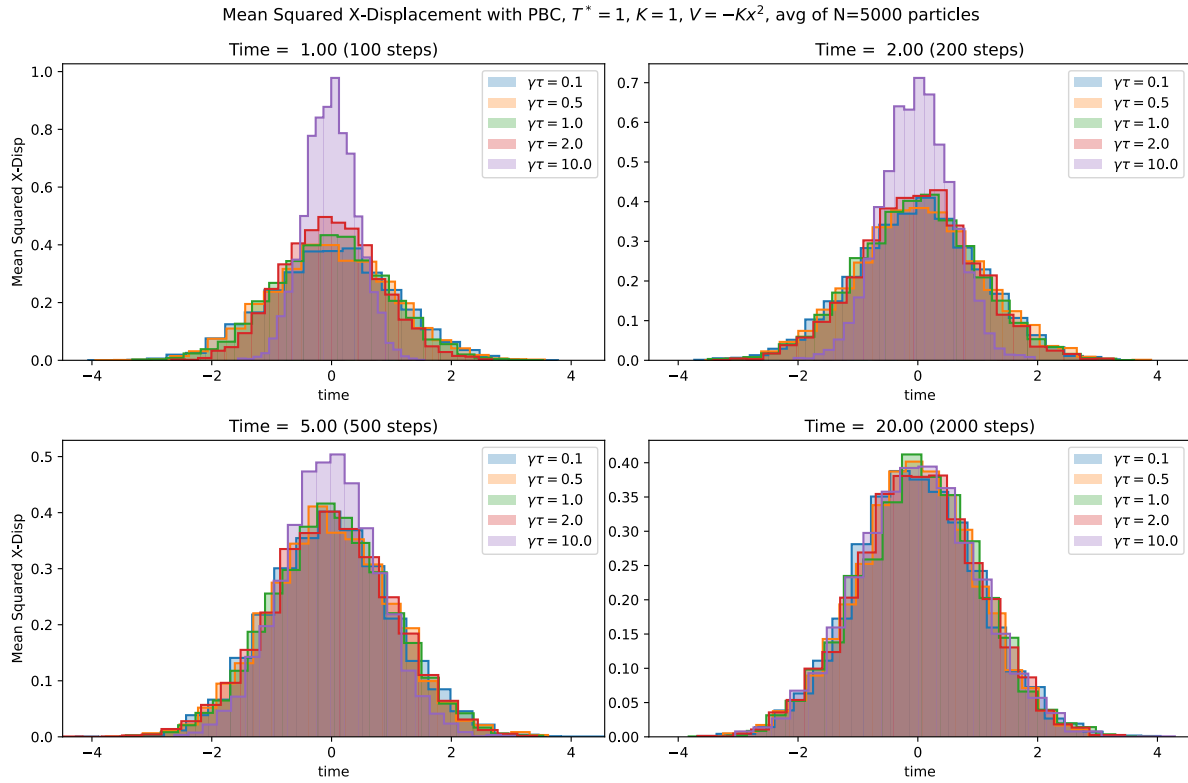
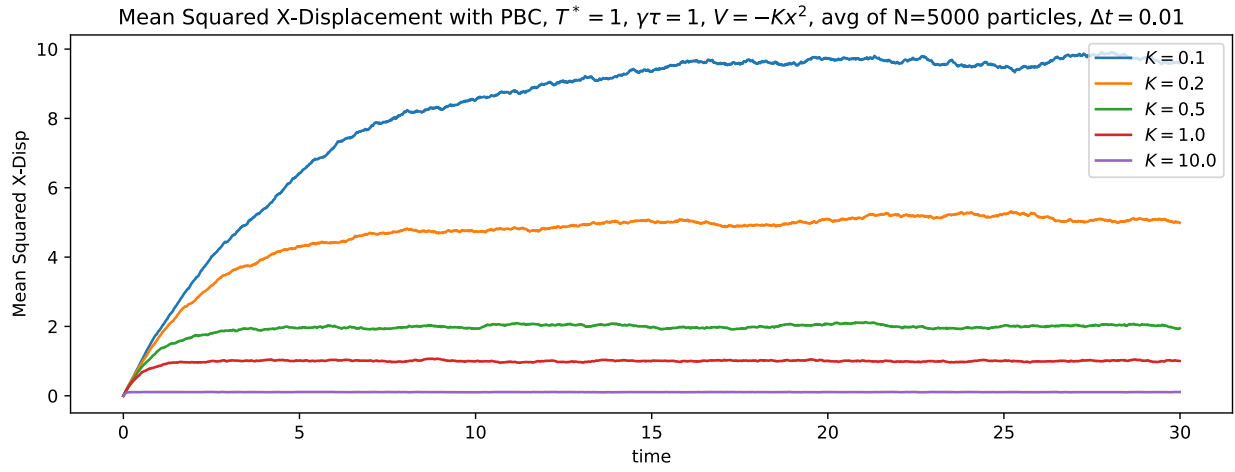
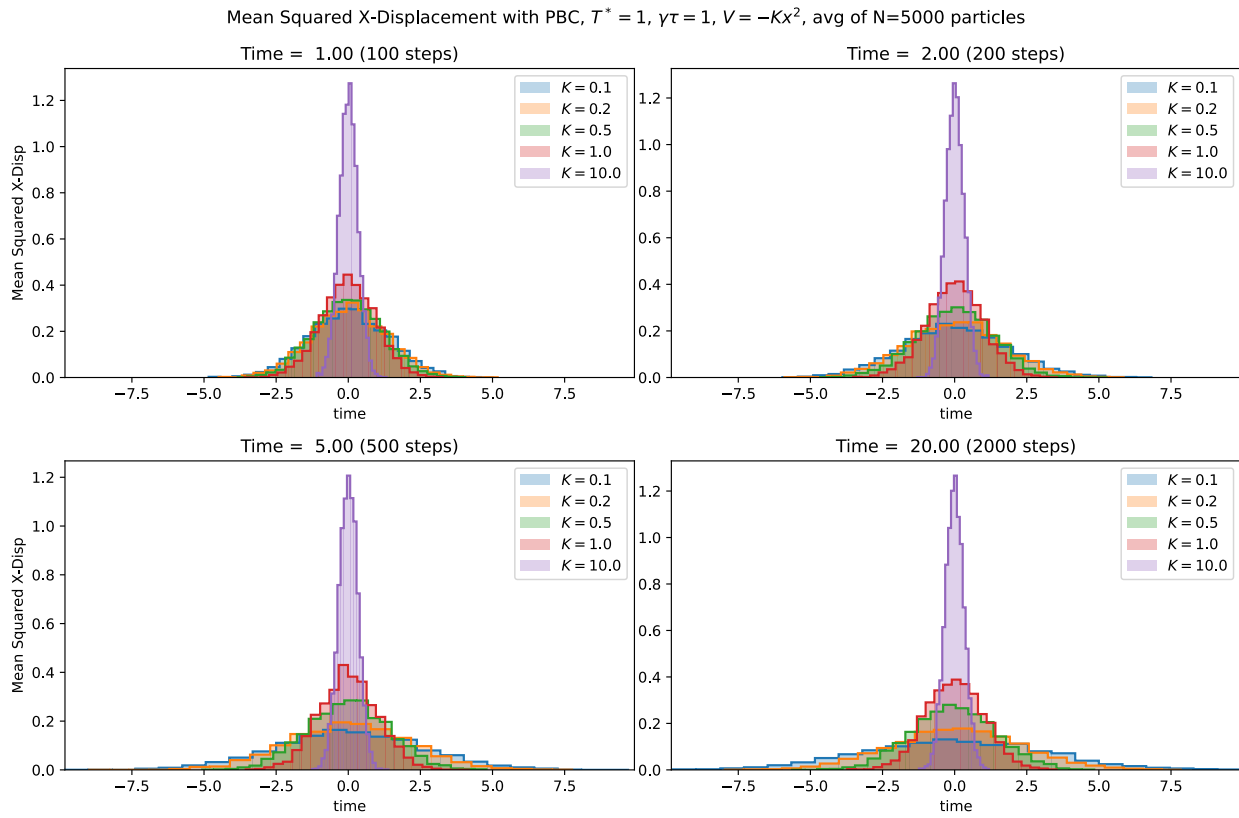


Figure 11.8: Distribution of the mean squared displacement for different $\gamma\tau$ s with T and K fixed for different times.

Figure 11.9: Mean squared displacement for different K s with T and $\gamma\tau$ fixed.Figure 11.10: Distribution of the mean squared displacement for different K s with T and $\gamma\tau$ fixed for different times.

Exercises Lecture 12

12.1 Exercise on the Multiple Histogram Method (MHM)

Choose a model with a phase transition. The standard choice is the grafted polymer model of Lesson 6. Otherwise, you should describe the model in the submitted notes. Temperature T , or better, its inverse β (with $k_B = 1$), is a parameter whose variation leads to a change in the phase of the system.

Implement the MHM to compute the energy variance per degree of freedom, $C(\beta) = \frac{1}{N}(\langle E^2 \rangle - \langle E \rangle^2)$ (proportional to the specific heat, $C_{sp} = \beta^2 C = N^{-1} \partial_T \langle E \rangle$). The following procedure assumes that the model follows a finite-size scaling close to the critical point at β_c ,

$$C(\beta) = N^{2\phi-1} f((\beta - \beta_c) N^\phi)$$

The scaling function f , in this case, has a maximum, i.e. a "peak".

For different system sizes N , compute $C_N(\beta)$ for a fine mesh of $\beta \in [0, \beta_{\max}]$ covering the finite-size proxy of the phase transition of the model, which is the peak of $C_N(\beta)$. Find peaks' position β_N and plot them as a function of $N^{-\phi}$ with crossover exponent ϕ to be determined by trying to align points in the $y = \beta_N$ vs $x = N^{-\phi}$ diagram. For the grafted polymers, a $\phi = 1/2$ should work fine.

Try to extrapolate the infinite-size limit β_c from the trend of β_N vs $N^{-\phi}$ points. This method has a single parameter ϕ . Including finite-size corrections amounts to fit points $y = \beta_N$ with $y(x(N)) = N^{-\phi}(1 + AN^{-\Delta}) = x(1 + Ax^{\phi\Delta})$ including a second parameter A .

12.1.1 Solution

The model used to implement the MHM is the Ising Model of Lesson 5. The model has periodic boundary condition and there is no external magnetic field. This model was chosen because the energy it's always an integer and it is usually decently spread (unless when we are too far to the left of T_c). Since the energy can take only integer numbers, it's easy to count the occurrences of all the energy values, and there is only one trivial way to make the binning. Once we have binned the energy, we use the histogram version of the formula of the MHM. The alternative would be to not compute any histogram, but then we would have to cycle over every sample of the energy when computing the quantities and the observables with this method. Usually you do this because you want to avoid choosing the binning, but in this case there is only one (trivial) binning! Also, the data can also be aggregated together and then the occurrences for each energy is calculated (automatically by the Python library). This speeds up the computation a lot, instead of cycling over M^{nsim} values, you cycle over less than thousand values (usually).

The formula used to calculate the Z_k (iteratively) is:

$$Z_k = \sum_E \frac{N(E)}{M \sum_j Z_j^{-1} e^{(\beta_k - \beta_j)E}} \quad (12.1)$$

and analogous equation were used to calculate $Z(\beta)$ and $O(\beta)$. The number of iteration M is equal for every simulation.

We noticed how important is to get the energy distribution overlapping a lot, otherwise a lot of over and under flow would appear when calculating all the parameters. This restrained the choices for the initial β s to be very close to each others, and thus close to β_c , since we are more interested in that region.

Our choice of L s, the sides of the grid, is $L = 20, 30, 40, \dots, 100$ and $N = L^2$. For each L we simulate the system with different 10 temperatures close to the theoretical critical temperature. All simulation are computed for 200000 steps, but the first 10000 points are discarded.

Then we proceed to calculate Z_k iteratively with the convergence criterion and the rescaling by:

$$A = \left[\min_k (Z_k) \max_k (Z_k) \right]^{-1/2} \quad (12.2)$$

After calculating Z_k we can calculate $Z(\beta)$ for a fine mesh of β s and

$$C^{\text{sp}}(\beta) = \frac{\beta^2}{N} (\langle E^2 \rangle - \langle E \rangle^2) \stackrel{MHM}{=} \frac{\beta^2}{N} (E^2(\beta) - (E(\beta))^2) \quad (12.3)$$

where

$$E^2(\beta) = \frac{1}{Z(\beta)} \sum_E \frac{E^2 \cdot N(E)}{M \sum_j Z_j^{-1} e^{(\beta_k - \beta_j)E}} \quad E(\beta) = \frac{1}{Z(\beta)} \sum_E \frac{E \cdot N(E)}{M \sum_j Z_j^{-1} e^{(\beta_k - \beta_j)E}} \quad (12.4)$$

and

$$Z(\beta) = \sum_E \frac{N(E)}{M \sum_j Z_j^{-1} e^{(\beta_k - \beta_j)E}} \quad (12.5)$$

We can also calculates C^{sp} without the MHM method (as seen in the middle of (12.3)) to compare them. First let's show the overlapping between then distribution of energy for the different T s. In figure 12.1 we can see the overlapping of three energy distribution for two values of L ; there is a good overlapping.

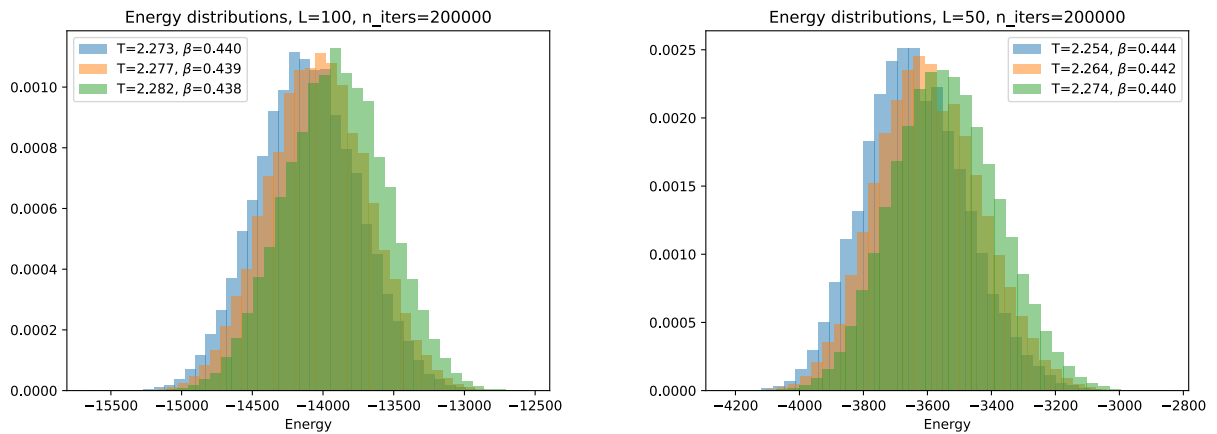


Figure 12.1: Overlapping distribution of the energy for two different L and three different temperatures.

Now let's plot $C^{\text{sp}}(\beta)$ for a fine mesh of β s and compare them with the one calculated directly with all the energies and the theoretical T_c . The graph is shown in figure 12.2, and the β s are converted in temperatures.

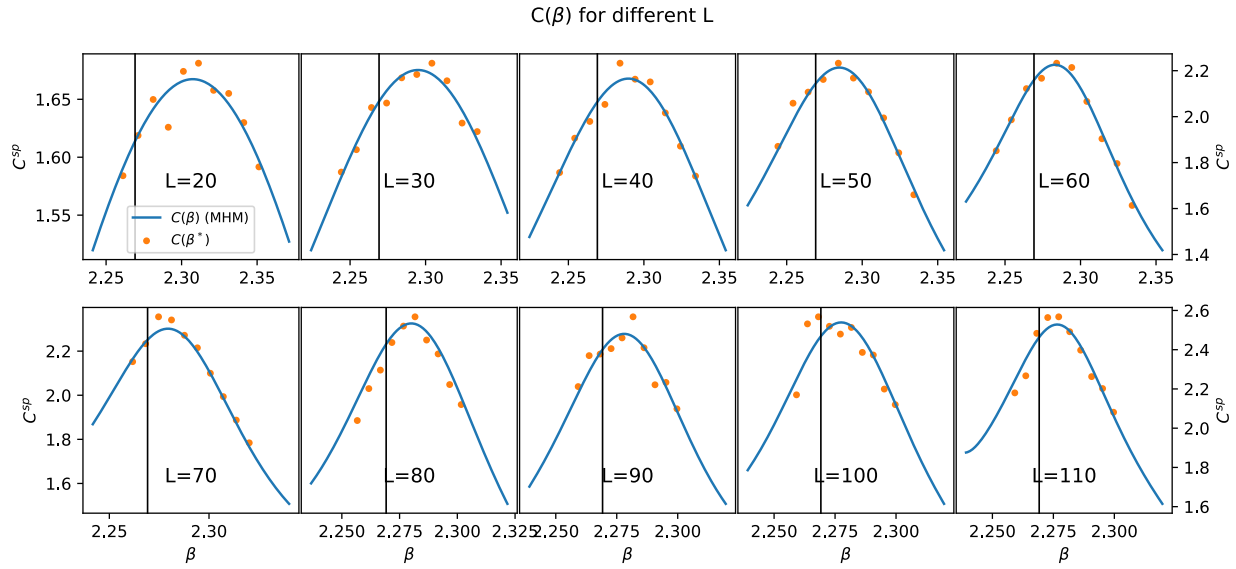


Figure 12.2: Plot of $C(\beta)$ for a fine mesh of β s around the interesting zone. In reality here are plotted the temperatures, so the reciprocal of β s. The dots are $C(\beta^*)$ calculated directly with the energies by the definition (without MHM), β^* are the temperatures used in the simulation. The vertical lines is the theoretical T_c for this model: $T_c = 2/\ln(1 + \sqrt{2})$.

It's easy to get the β where the curve is at the maximum, since the peak is very clear. With all the β_{peak} and the respective N , we calculate the crossover exponent ϕ . We have a range of possible ϕ s, and for each we calculate the linear regression for $x = N^{-\phi}$ and $y = \beta_{\text{peak}}$. Then we calculate the y-residuals squared as:

$$Res = \sum_i (y_i - (mx_i + q))^2 \quad (12.6)$$

and we choose the line with the lowest residuals. By doing that we find the crossover exponent $\phi = 0.915$ and the intercept $q = 0.441$ (figure 12.3). The latter tells us the value of $\beta_{c, L=\infty}$ assuming the trend given by the line holds. $\beta_{c, L=\infty}$ is the critical temperature (its inverse) for a system with $L \rightarrow \infty$, $T_{c, L=\infty} = 2.268$, which compared to theoretical one $T_c = 2/\ln(1 + \sqrt{2}) = 2.269$ is in good agreement.

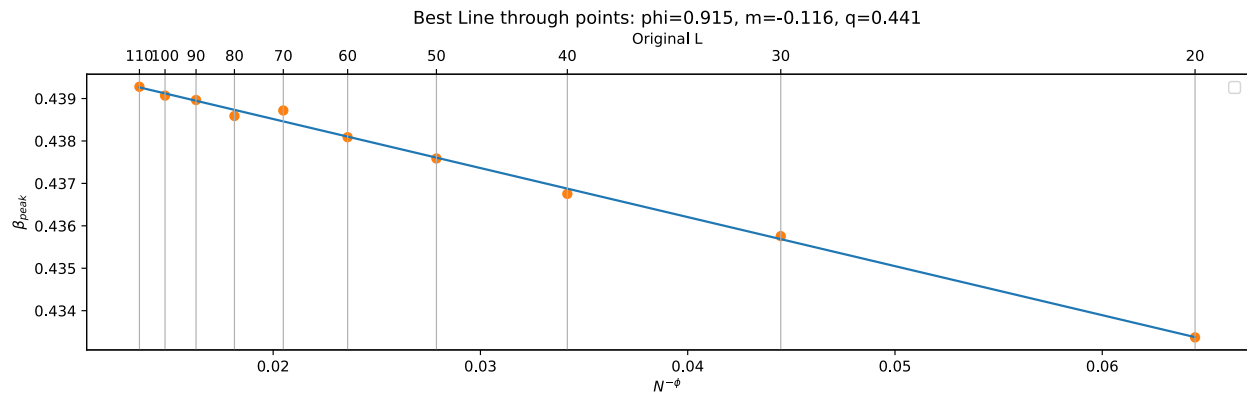


Figure 12.3: Best ϕ which makes the best line passing through the points β_{peak} .

Exercises Lecture 13

13.1 Exercise on the cell list

Simulate N Brownian particles in a two-dimensional box of size $B \times B$, with reflective boundary conditions. Each particle has position $\vec{x}_i = (x_i^0, x_i^1)$ and follows the time-discretized Euler-Maruyama dynamics with increments

$$d\vec{x}_i = \mu_i \vec{F}_i(x) dt + \sqrt{2D_i} \vec{B}_i$$

with time step dt , mobility μ_i , diffusion constant D_i , force $\vec{F}_i(x)$ depending on all particles' positions $x = (\vec{x}_0, \dots, \vec{x}_{N-1})$. Each component of the Gaussian white noise term $\mathcal{B}_i^\alpha$ ($\alpha = 0, 1$) is drawn from a zero-mean normal distribution with variance $\sqrt{dt}$.

Each pair of particles interacts with a repulsive potential energy depending on their distance $r_{ij} = |\vec{x}_i - \vec{x}_j|$ and a reference distance σ_{ij} :

$$U(r_{ij}) = \varepsilon e^{-r_{ij}^2 / (2\sigma_{ij}^2)}$$

We assign a radius R_i to each particle and set $\sigma_{ij} = R_i + R_j$. Note that $U(|\vec{r}|)$ yields a central force $\vec{F}_U(|\vec{r}|)\hat{r}$. We cutoff this force at $r_c = 4\sigma_{ij}$ (i.e., it is set to zero for $r > r_c$). The total force F_i on each particle is a sum of F_U pairwise forces with other particles in its surroundings.

To introduce thermodynamics and physics at a temperature T (with $k_B = 1$), we set $D_i = \mu_i T$, and mimic Stokes' law by setting $\mu_i = 1/R_i$.

The goal of this exercise is simulating this system also for large N values. To this end, implement the cell list algorithm.

With everything in place, study the diffusivity as a function of the number density $\rho = N/B^2$. Specifically, study the mean square displacement along the x-direction, $\Delta(t) = \langle [x^0(t) - x^0(0)]^2 \rangle$ for different ρ values. Usually, such quantity is plotted as a function of t in a log-log diagram. For low ρ , the mean square displacement should follow $2D_i t$ up to a saturation due to the finite box size. For large ρ , the behavior should be more complex, effectively with lower diffusivity and possibly with plateaus due to particles' caging. Such effect emerges if neighboring particles do not move much due to the high density and constrain the observed particle to remain caged for some time in a region. For large ρ , the system would crystallize if all R_i are the same. We avoid this by choosing $R_i = R^A$ for half of the particles and $R_i = R^B \neq R^A$ for the other ones.

Parameters:

- $B = 100$
- $R_i = R^A = 1.25$ for $0 \leq i < N/2$ (particles of kind A)
- $R_i = R^B = 1$ for $N/2 \leq i < N$ (particles of kind B)
- $T = 1$

- $\varepsilon = 10$
 - A possible list of N to have ρ over some orders of magnitude is $N = 2, 10, 50, 200$ and maybe 500.
1. Which side ℓ for the cell in the cell list do you choose with these parameters?
 2. Is $\Delta^A(t)$ for particles A different from $\Delta^B(t)$ for particles B?
 3. How do Δ 's change with ρ ?

13.1.1 Solution

The implementation of our algorithm for cell list is the same as seen during the lesson, but when we actually use it to calculate the force we approach the iteration in a different way. One could, for each particle, find the candidate neighbors with the cell list structure, calculate the distance, check if it is under the cutoff, calculate the force and then proceed to the next particle. Instead we select the first cell c_1 , then for every of its neighbor cells c_2 we do all the comparison between the particles p_i inside c_1 with the particles p_j inside c_2 . Then we move to the next cell and repeat the process. Also, since we don't want to recalculate things, we can visit only the lower right part (4 out of 8 and the cell itself, visiting 5 cells in total) in the same way with the Ising model we visit only two nearest neighbors out of four with the condition $i < j$. In figure 13.1 there is an example of which neighbors every cell visits.

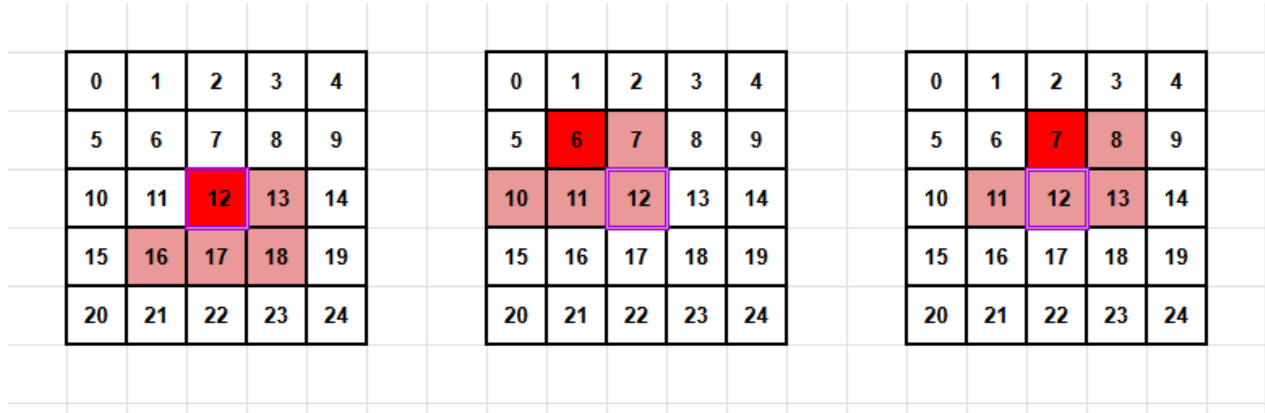


Figure 13.1: Example of 5x5 cell system. For each particles inside cell 12, we compare them with all the particles inside cells 13, 16, 17 and 18. We don't compare them with particles in cells 6, 7, 8 and 11 because we already did those comparison when we were at those cells in the previous steps.

All of this makes sense because we have an array f_i of the force each particle p_i experiences. So every time we make a comparison between p_i and p_j we calculate F_{ij} and we add it in f_i , but also we add $-F_{ij} = F_{ji}$ in f_j . At the end of the cells iteration, each f_i will contain the sum of all the forces exerting on p_i . After calculating the forces, we then update the positions using those forces.

The choice of the side ℓ for the cell was made such that all the possible particles interacting with a cell are only the particles in the neighbors cells (and the cell itself). Since the biggest interaction is for particles of type A, with a $\sigma_{AA} = 4 \cdot 2 \cdot R^A = 10$, the cell side need to be at least 10. Since the side of the box is 100, then was trivial to choose 100 cells in total with side $\ell = 10$.

About the initial condition, the particles are arranged in a square grid with spacing $\lceil \sqrt{N} \rceil$. They type is instead assigned at random (keeping the 50% - 50% ratio). In figure 13.2 are shown the initial configuration for $N = 200$ and $N = 500$.

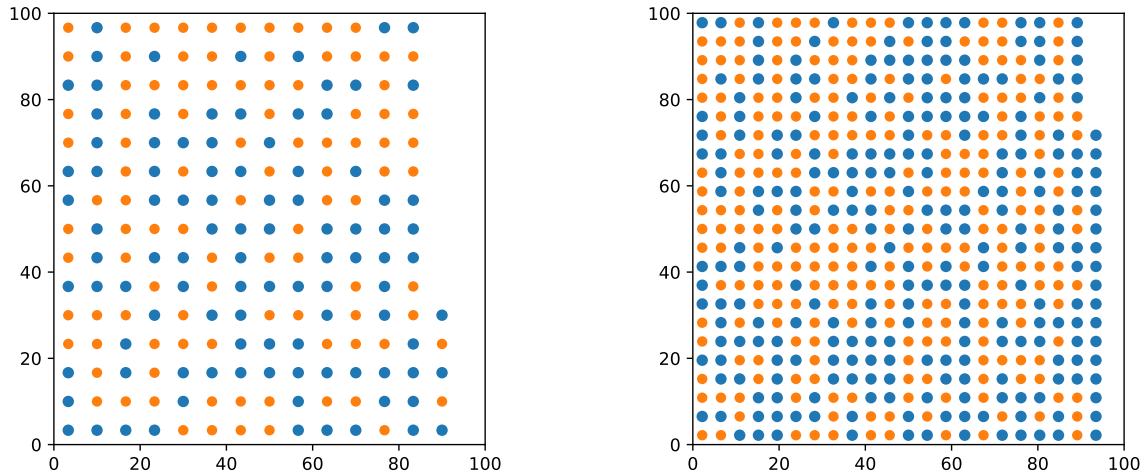


Figure 13.2: Initial configuration for $N = 200$ and $N = 500$ respectively. The types are assigned at random, given N the positions are fixed. The radius of the particles is not to scale with the box, but the relative radius between the two types is correct.

The simulation is done for 200000 iterations with $\Delta t = 0.2$ for $N = 30, 50, 100, 200, 500$. The total time for these 5 simulations is ≈ 65 seconds with a raw binary output of 2.63GB (all the positions at each time are stored).

In figure 13.3 is shown how the Δ changes (for the x-position of the particles) for different ρ 's ($\rho = N/V$) and for the two different types of particles as a function of time. Note that this plot had some processing applied to make it (more) readable. Since the time increases linearly, the points on a log-log graph tend to accumulate on the right. To avoid having half of the points in a fraction of the image, the following was applied. First the image was divided in 500 columns of equal size, then the points inside were averaged to produce one point. This was done only if the column had more than 3 points inside (practically, this means that the method was applied only after a certain x).

The only big difference is seen with $N = 500$ where the mean square displacement increase very rapidly until it reach the equilibrium value. Probably this happens because of the initial configuration of the grid, where all the particles are too close to each others, and thus receiving bigger push on average. Can also be seen that the smaller particles (type B) have consistently smaller MSD, but if we look at the others simulation this is not true.

Three animations are available [here](#) on GitHub.

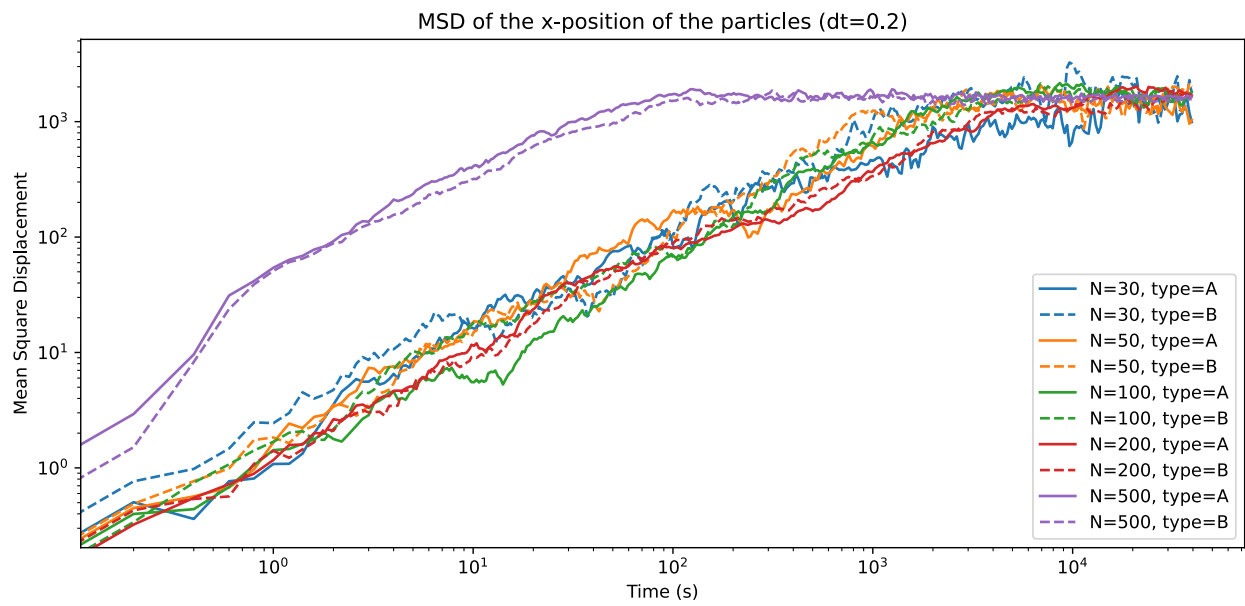


Figure 13.3: Plot of the MSD of the x-position of the particles for different N and for the two types. Some processing was applied to the graph to avoid the accumulation of the points to the right, as written in the text.